

Acute Activation of Genes Through Transcriptional Condensates Impact Non-target Genes in a Chromatin Domain

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The authors use single molecule imaging and in vivo loop-capture genomic approaches to investigate estrogen mediated enhancer-target gene activation in human cancer cells. These potentially **important** results suggest that ER-alpha can, in a temporal delay, activate a non-target gene *TFF3*, which is in proximity to the main target gene *TFF1*, even though the estrogen responsive enhancer does not loop with the *TFF3* promoter. To explain these results, the authors invoke a transcriptional condensate model. The claim of a temporal delay and effects of the target gene transcription on the non-target gene expression are supported by **solid** evidence but there is no direct evidence of the role of a condensate in mediating this effect. The reviewers appreciate that the authors have done a lot of work to strengthen the study. This work will be of interest to those studying transcriptional gene regulation and hormone-aggravated cancers.

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Abstract

Transcription activation of genes by estrogen is driven by enhancers, which are often located within the same Topologically Associating Domain (TAD) as non-targeted promoters. We investigated how acute enhancer-driven activation affects neighbouring non-target genes within the same TAD. Using single-molecule RNA FISH (smFISH), we tracked the transcription of *TFF1* (enhancer-targeted) and *TFF3* (non-targeted) during estrogen stimulation. We observed mutually exclusive expression patterns: *TFF1* expression peaked at 1 hour, while *TFF3* reached its peak at 3 hours, after *TFF1*'s activation had diminished. Chromatin looping data indicated that the enhancer loops with *TFF1* but not *TFF3*, suggesting that *TFF3* upregulation is not due to direct enhancer-promoter interactions. CRISPR deletion of the *TFF1* enhancer and 1,6-hexanediol (HD) exposure revealed that the *TFF1* enhancer:promoter undergo Liquid-Liquid Phase Separation (LLPS), which sequesters the transcriptional machinery and inhibits *TFF3* expression. As estrogen signalling wanes or LLPS is disrupted,

TFF1 expression declines while *TFF3* expression increases. Our findings reveal that enhancer-driven activation can indirectly influence neighbouring genes, highlighting a dynamic shift in gene expression as signalling progresses.

Introduction

Acute transcriptional activation of ligand-induced genes drives downstream signalling response. Similar to development-specific genes, signalling induced genes are also driven by enhancers (Hah et al., 2013 [↗](#); Li et al., 2013 [↗](#); Liu et al., 2014 [↗](#); Uyehara & Apostolou, 2023 [↗](#)). Upon binding with ligand-induced transcription factors (TFs), these enhancers loop with their target promoters mostly, in the same TAD (Buecker et al., 2014 [↗](#); Bulger & Groudine, 1999 [↗](#); Chepelev et al., 2012 [↗](#); Furlong & Levine, 2018 [↗](#); Oh et al., 2021 [↗](#); Panigrahi & O'Malley, 2021 [↗](#); Ptashne, 1986 [↗](#); Sanyal et al., 2012 [↗](#); Yan et al., 2018 [↗](#)). Enhancer: promoter pairing is thought to be specific and forms the basis of noise-free gene activation of a subset of genes crucial for signalling response (Bojcsuk et al., 2017 [↗](#); H. Chen et al., 2018 [↗](#); Friedman et al., 2024 [↗](#); Galouzis & Furlong, 2022 [↗](#); Zabidi et al., 2015 [↗](#)). Due to such specificity, gene transcription occurs in waves of early and late responsive genes (Fowler et al., 2011 [↗](#); Yamamoto & Alberts, 1976 [↗](#)). Often, the protein factors translated from early genes regulate the expression of late responsive genes (Dixon et al., 1996 [↗](#); Freter et al., 1996 [↗](#); Herschman, 1991 [↗](#); Williams et al., 1999 [↗](#); Winkles, 1997 [↗](#); Winston & Pledger, 1993 [↗](#)). However, it is not known if the genes that are activated early in the signalling time course are spatially related to late activating genes. Further, if the spatial proximity of any gene to an early gene is enough to cause the temporal differential transcription due to sequestration of the transcriptional machinery from late gene to early gene is poorly understood. Transcription factors (TFs), after binding to cognate DNA motifs, recruit RNA polymerase machinery and co-activators for gene activation (Levine & Tjian, 2003 [↗](#); Liu et al., 2014 [↗](#)). These machinery are limited in supply, and may become sequestered from other genomic regions (Koşar & Erbaş, 2022 [↗](#)). Further, acute activation of genes is linked with phase-separation of TFs, polymerases, mediators and other co-factors/activators, most of which harbour low complexity regions to promote weak protein-protein interactions driving the formation of TF-condensates (Boehning et al., 2018 [↗](#); Boija et al., 2018 [↗](#); Cai et al., 2019 [↗](#); L. Chen et al., 2023 [↗](#); Cho et al., 2018 [↗](#); Chong et al., 2018 [↗](#), 2022 [↗](#); Mann & Notani, 2023 [↗](#); Sabari et al., 2018 [↗](#); Shrinivas et al., 2019 [↗](#); Stortz et al., 2020 [↗](#), 2024 [↗](#)). Though the precise stoichiometry of protein molecules in these condensates is unknown, such structures involve multiple molecules of each transcriptional protein. The formation of such phase-separated compartments can potentially act as a sink for transcriptional machinery depriving neighbouring promoters and enhancers that are not part of the compartment. Such sequestration would cause indirect suppression of these spatially proximal genes in the same TAD but not the part of the same condensate.

In order to investigate the effect of an enhancer: promoter pair on a neighbouring gene within the same TAD, we looked at the paradigmatic model of estrogen signalling in mammary epithelial cells, namely MCF7 (Levenson & Jordan, 1997 [↗](#); Masiakowski et al., 1982 [↗](#)). E2-signalling can be induced within minutes by treating the cells with estradiol (E2), which causes activation and repression of several genes across the genome. The peak of E2-mediated signalling occurs 40 minutes post-induction and starts to decay by 160 minutes (Hah et al., 2011 [↗](#)). This occurs via the binding of estrogen receptor-alpha (ERα) on the enhancers (Li et al., 2013 [↗](#)). We selected E2-induced E-P pair *TFF1* and its neighbouring gene *TFF3* on Chr21 as discussed below (Chinery et al., 1996 [↗](#)). Several studies have used single-cell methods to interrogate E2 responsive genes such as *GREB1*, *MYC* and *TFF1* as a paradigm to understand their transcriptional response upon estrogen stimulation (Fritzsch et al., 2018; Patange et al., 2022 [↗](#); Rodriguez et al., 2019 [↗](#); Stossi et al., 2020 [↗](#)). These studies have implicated the cellular states as determinants of transcriptional response and that the long repressive states of *TFF1* give rise to expression variability in isogenic lines (Rodriguez et al., 2019 [↗](#)).

TFF1 is found in a topologically associated domain (TAD) along with a few other genes, including *TFF2*, *TFF3*, *TMPRSS3*, and *UBASH3A* (Oh et al., 2021 [↗](#); Quintin et al., 2014 [↗](#); Rao et al., 2014 [↗](#)). *TFF1* and *TFF3* is located at 10kb and 60kb from enhancer, respectively (Fig. 1A [↗](#)), and this locus has been useful in answering the questions about promoter-enhancer interactions and multi-gene regulation (Oh et al., 2021 [↗](#); Quintin et al., 2014 [↗](#)).

We chose to answer these questions using an integrated approach consisting of single molecule RNA FISH (smFISH), genome-wide conformation capture and sequencing (4C-seq), and perturbation of cis-acting regulatory elements by CRISPR. We identified that *TFF1* and *TFF3* genes located within the same TAD show distinct and opposing expression profiles over the course of E2-mediated signalling. In agreement with this, the enhancer also showed increased looping interaction with the cognate gene promoter during peak expression at 1h of signalling. We also identified the role of TF-mediated LLPS on the regulation of these two genes during signalling. Hence, we propose that ligand-dependent condensation of ER α can support the expression of the cognate gene in concert with the enhancer while negatively affecting the expression of a neighbouring gene.

Results

ER α binding and ligand-induced gene expression of *TFF1* and *TFF3* change over the course of estrogen signaling

In order to understand how acute activation of one gene in a TAD affects the transcription of a proximate gene, we chose to study *TFF1* and its neighbouring gene *TFF3* at a 43kb distance within same TAD. *TFF1* expression is linked with an enhancer located 10 kb downstream (Li et al., 2013 [↗](#); Saravanan et al., 2020 [↗](#); Oh et al., 2021 [↗](#)) and the distance between *TFF1* enhancer and *TFF3* gene is 53kb (Fig. 1A [↗](#)).

17- β -oestradiol exposure leads to ER α binding on regulatory regions, leading to gene activation. Gene transcription of E2-regulated genes was shown to peak at 1h and significantly reduced at 3h due to the rapid degradation of ER α (Hah et al., 2011 [↗](#)). The binding of ER α in the genome also follows this temporal kinetics, where it peaks at 1h and reduces at 3h post ligand stimulation (Hah et al., 2011 [↗](#); Li et al., 2013 [↗](#); Liu et al., 2014 [↗](#)). At the *TFF1* locus, the *TFF1* enhancer and to some extent, its promoter, were bound by ER α even in the absence of E2 induction, while it increased in strength at these regions and also at various other regions in the locus at 1h of ligand stimulation (Saravanan et al., 2020 [↗](#)). Notably, the binding at these regions substantially decreased at 3h. On the other hand, ER α did not bind on the *TFF3* promoter throughout the course of signalling (Fig. 1A [↗](#) and Fig. S1B [↗](#)). A single peak of ER α was observed approx. 3 kb upstream to *TFF3* gene. We have previously shown that estrogen-induced clustered binding of ER α on *TFF1* region is associated with its acute activation post estrogen stimulation (Saravanan et al., 2020 [↗](#)). We then tested their expression level by nascent RNA-seq (GRO-seq). We observed a dramatic transcriptional activation of *TFF1* at 1h, which reduced substantially at 3h (Fig. 1A [↗](#)). The expression level of *TFF3* was comparatively lower and exhibited mild induction at 1h which increased furthermore at 3h upon signalling (Fig. 1A [↗](#), Fig. S1C [↗](#)). However, qRT-PCR on unspliced *TFF3* exhibited upregulation at 1h and down-regulation at 3h (Fig. 1B [↗](#)). We reasoned that these differences in *TFF3* could be because of its low baseline expression (Clark et al., 2015 [↗](#); Conesa et al., 2016 [↗](#); Sha et al., 2015 [↗](#); Svensson et al., 2017 [↗](#); Tarazona et al., 2011 [↗](#)), and lower yet expression of unspliced transcripts. Therefore, in order to test the relative induction of these genes and their co-regulation, we decided to perform single-cell measurements of RNA (Fig. S1A [↗](#)) that do not rely on PCR amplification and can provide absolute numbers of *TFF1* and *TFF3* transcripts in a given cell. Towards this, we chose to employ smFISH (Femino et al., 1998 [↗](#); Haimovich & Gerst, 2018 [↗](#); Kwon, 2013 [↗](#); Raj et al., 2008 [↗](#)).

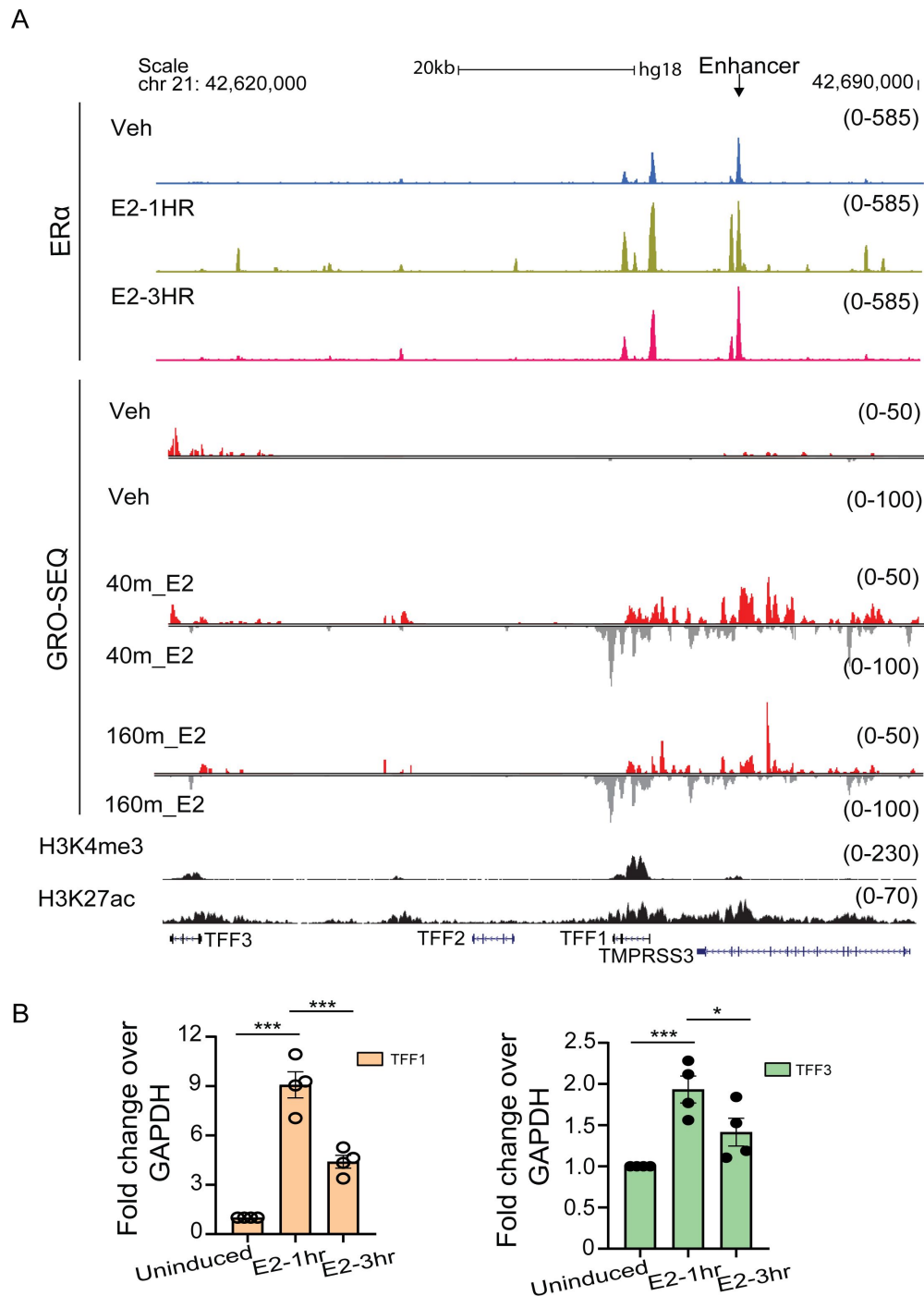


Figure 1.

ERα binding and ligand-induced gene expression of *TFF1* and *TFF3* change over the course of estrogen signaling

A. Schematic depicting TFF1 locus, UCSC genome browser snapshots showing the binding of ERα, H3K27ac status, H3K4me3 signal, and Gro-seq signal for robustly E2-induced TFF1 locus. First, second and third ERα ChIP-seq and Gro-seq tracks are from vehicle-treated, E2-1h and E2-3h in WT cells, respectively.

B. qRT-PCR showing the changes in expression of TFF1 and TFF3 genes during the E2 signaling time course. Error bars denote SEM from four biological replicates. Each dot represents a replicate. p-values were calculated by Student's two-tailed unpaired t-test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

***TFF1* and *TFF3* exhibit opposite trends during the E2 signalling time-course**

MCF7 cells are hypertriploid to hypotetraploid, and each of the three alleles within a nucleus can behave differently, and smFISH is pre-eminently suited to capturing cell- and allele-specific heterogeneity of expression even for low expressing genes, compared to bulk studies that average over cell populations. Briefly, smFISH allows to visualize single RNA molecules in fixed cells using multiple fluorescently labelled oligonucleotide probes targeted to the RNA of interest and therefore, it can be used to image the transcription and the localization of multiple gene transcripts at same and different time points after signalling (Femino et al., 1998 [DOI](#); Raj et al., 2008 [DOI](#)).

We designed the smFISH probes targeting the intronic region of *TFF1* and *TFF3* to measure their nascent transcripts (referred to as InTFF1 and InTFF3 hereon) whereas the exonic probes were used to primarily measure mature mRNA transcripts (referred to as ExTFF1 and ExTFF3 hereon). Intronic probes are particularly suited for investigating nascent transcriptional status at the time of fixation, while probes against mature mRNA reflects on more steady-state levels of functional mRNA due to finite mRNA lifetimes (Skinner et al., 2016 [DOI](#)). Additionally, the intronic probes can help determine the localisation and number of alleles that are transcribing at a given time thus allowing for additional interpretations regarding the expression of multiple genes (Skinner et al., 2016 [DOI](#)).

Representative images from the smFISH experiment probing InTFF1 and InTFF3 are depicted (**Fig. 2A** [DOI](#)). The smaller foci represent individual intronic/nascent transcripts while the larger foci represent the site of transcription (Raj et al., 2006 [DOI](#); Zenklusen et al., 2008 [DOI](#)). Since MCF7 cells are hypertriploid in nature, we expected to see 1-3 large foci per cell representing sites of transcription. Additionally, we observed many individual transcripts labelled by intronic probes (**Fig. 2A, D** [DOI](#)) within the same nuclei (probes against mature RNA are usually more cytoplasmic). This is suggestive of the fact that the transcripts undergo non-co-transcriptional splicing as the transcripts labelled by intronic probes are localised away from the site of transcription. This is not surprising as several reports have shown that nascent transcripts can undergo splicing well after transcription (Coulon et al., 2014 [DOI](#); Drexler et al., 2020 [DOI](#); Khodor et al., 2012 [DOI](#)).

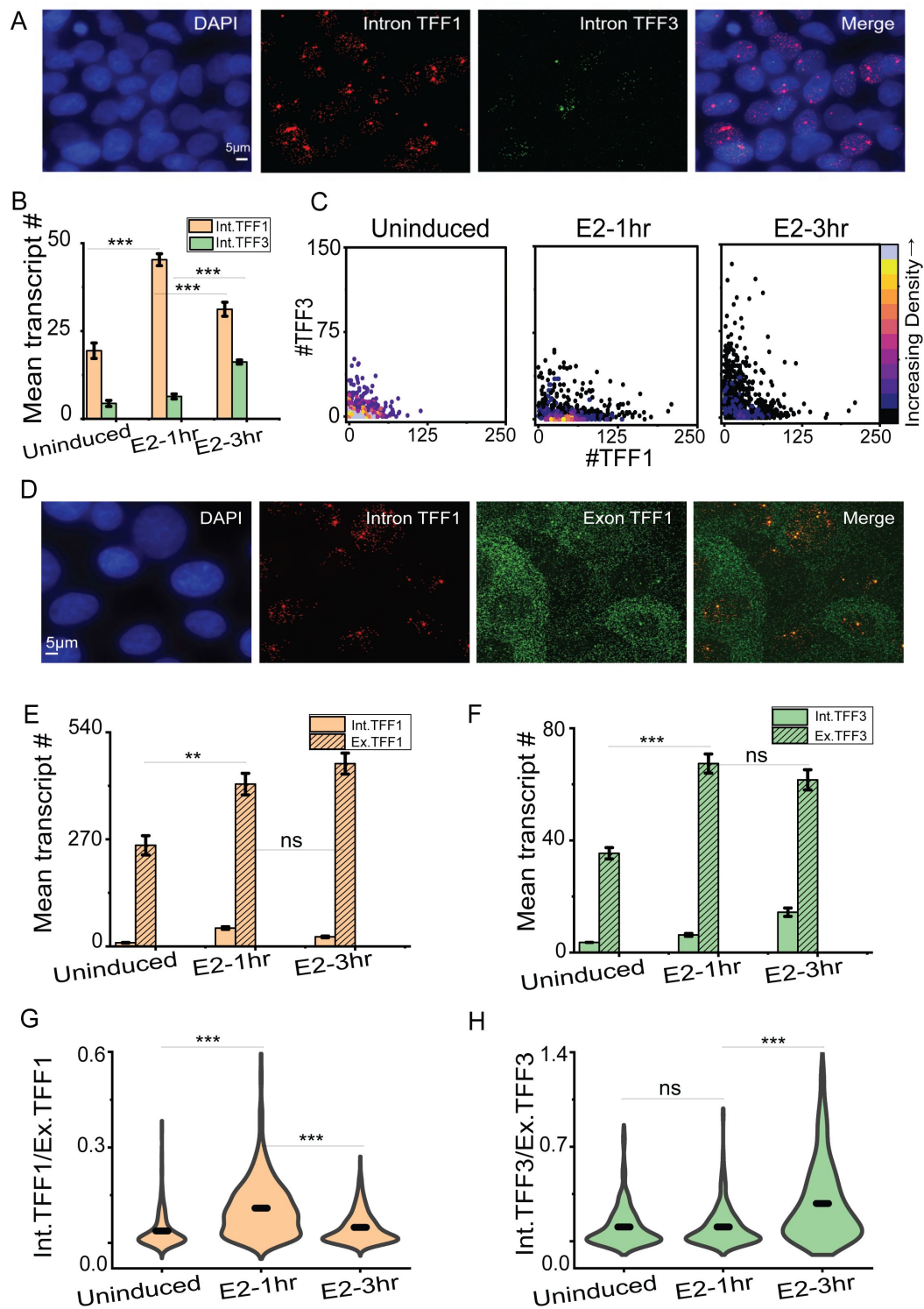


Figure 2.

TFF1 and TFF3 expressions show opposite trends during the E2 signaling time-course

A. 60X Representative images from single molecule RNA FISH experiment showing transcripts for TFF1 and TFF3. The probe was designed against the unsplined RNA containing the intronic region. The scale bar is 5 microns.

B. The mean RNA numbers are depicted. These are counted using an in-house MATLAB code which uses the DAPI-stained nuclei as the mask to count the RNA present in the nucleus. The graph shows the mean of means from three different repeats of the experiment, and error bars denote SEM ($n = 665$, $N = 3$). p-values were calculated by Student's two-tailed unpaired t-test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

C. Scatter plots showing the distribution of InTFF1 and InTFF3 on a cell-by-cell basis ($n = 665$, $N = 3$). The absolute RNA numbers are combined from three different repeats. Density plots have been used to clearly visualize overlapping data points.

D. 60X Representative images from single molecule RNA FISH experiment showing transcripts for InTFF1 and ExTFF1. Scale bar is 5 microns.

E. The mean RNA numbers for InTFF1 and ExTFF1 are depicted. Separate probes were used to target unspliced (InTFF1) and mature (ExTFF1) RNA. These are counted using an in-house MATLAB code which uses the DAPI-stained nuclei as the mask to count the intronic RNA present in the nucleus and a free-drawn region to designate the cell to count the exonic RNA present in the nucleus as well as the cytoplasm. The graph shows the mean of means from three different repeats of the experiment, and error bars denote SEM ($n > 360$, $N = 3$). p-values were calculated by Student's two-tailed unpaired t-test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

F. The mean RNA numbers for InTFF3 and ExTFF3 are depicted. Separate probes were used to target unspliced (InTFF3) and mature (ExTFF3) RNA. The graph shows the mean of means from three different repeats of the experiment, and error bars denote SEM ($n > 210$, $N = 3$). p-values were calculated by Student's two-tailed unpaired t-test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

G. Violin plots showing the ratio of intronic to exonic TFF1 counts are depicted. The graph shows the distribution of ratios combined from three different repeats ($n > 360$, $N = 3$). p-values were calculated by the Mann-Whitney test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

H. Violin plots showing the ratio of intronic to exonic TFF3 counts are depicted. The graph shows the distribution of ratios combined from three different repeats ($n > 210$, $N = 3$). p-values were calculated by the Mann-Whitney test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

We quantified the number of such transcripts per nuclei as a proxy for ongoing transcription. At 1h post E2 induction, *TFF1* mean transcript counts increased significantly compared to uninduced, whereas, the increment was less pronounced for *TFF3* transcripts. In contrast, the mean transcript counts for *TFF3* increased significantly at 3h post induction while *TFF1* transcription showed a decrease at 3h compared to 1h (**Fig. 2B**). We observed an increase in the number of transcribing sites per cell for both TFF1 and TFF3, as well as in the number of cells showing transcripts compared to the untreated condition (**Fig. 2B**, **Fig. S2A-B**). Further, TFF1 transcripts spots were more spread out in the nucleus that harboured multiple transcribing sites (**Fig. S2C-D**). To check the transcriptional status of *TFF1* and *TFF3* in the same cell, we plotted the transcript counts from individual cells (**Fig. 2C**). The data suggested that the transcript counts for *TFF1* increased at 1h and cells that showed high counts for *TFF1* were likely to have low counts for *TFF3* and vice-versa. But the RNA counts for *TFF1* decreased at 3h and increased for *TFF3*. Overall, it was evident that the transcriptional profile for these two genes located in the same TAD were negatively correlated as they peaked at different time points. To further confirm this, smFISH using probes targeting both the intronic and exonic transcripts in the same experiment was conducted. Intronic probes represent active transcription, while exonic probes show accumulated mature RNA from past transcriptional events even in the absence of active transcription (Skinner et al., 2016). Representative images from the smFISH experiment probing InTFF1/ExTFF1 and InTFF3/ExTFF3

are depicted (**Fig. 2D** [↗](#), **Fig. S2E** [↗](#)). InTFF1 spots overlapped with ExTFF1 in the nucleus (**Fig. S2F** [↗](#)) and InTFF1 spots were restricted to the nucleus, whereas ExTFF1 were both in the nucleus and the cytoplasm as expected, indicating the specificity of smFISH probes (**Fig. S2E** [↗](#)). We observed that the exonic transcript counts for both *TFF1* and *TFF3* increased at 1h compared to uninduced (**Fig. 2E, F** [↗](#)). Strikingly, at 3h, the exonic transcripts for *TFF1* continued to increase even while the intronic transcript counts reduced, though the fold increase (1.12 ± 0.02) was less compared to that between uninduced and 1h (1.63 ± 0.28), indicating a plateauing of steady state levels. Exonic transcripts for *TFF3* at 3h remained comparable to 1h while the intronic counts increased suggesting, *TFF1* transcription reduces at 3h, whereas *TFF3* expression increases. We reasoned that the ratio of intronic transcript number (InTFF) to exonic transcripts number (ExTFF) should represent the status of transcription as an increase in the number of intronic transcript due to expression would result in a higher ratio while also taking into consideration the number of mature transcripts. As expected, the ratio of intronic transcripts to exonic transcripts also showed that transcription is active at 1h for *TFF1* as the ratio is higher compared to uninduced and 3h (**Fig. 2G** [↗](#) and **Fig. S3A** [↗](#)). Contrastingly, the ratio was highest at 3h for *TFF3* suggesting that active transcription takes place much after the peak of E2-mediated signalling and maximal *TFF1* expression (**Fig. 2H** [↗](#) and **Fig. S3B** [↗](#)). Indeed, a very recent study has shown that such post-transcriptional splicing can occur for genes and intron dispersal can be expected more commonly for some highly expressed genes (Coté et al., 2023 [↗](#)). Therefore, the observation of intronic signal away from the site of transcription as we see for *TFF1* and *TFF3* is not unexpected (**Fig. 2A** [↗](#), **D, S2** [↗](#)).

***TFF1* enhancer does not change target promoters during signalling time-course**

In order to identify the molecular players behind the differential expression of these two genes that are in the same TAD, we asked if the ER α -bound enhancer downstream to the *TFF1* gene loops with *TFF1* at 1h, and with *TFF3* at 3h. This enhancer acutely activates *TFF1* at 1h post estrogen stimulation (Saravanan et al., 2020 [↗](#); Ho et al., 2021). We interrogated the looping using enhancer as a viewpoint by 4C-seq in uninduced, 1h and 3h post E2 stimulation. We observed robust interactions between enhancer and *TFF1* promoter at 1h post induction, which reduced at 3h. On the other hand, its interaction with the *TFF3* promoter exhibited very low counts in uninduced as well as E2-induced conditions at both time points and in both replicates (**Fig. 3A** [↗](#), **Fig. S4** [↗](#)). This suggests that the looping of enhancer potentially induces the expression of *TFF1* gene at 1h, and loss of interactions at 3h results in weak *TFF1* transcription. However, the lack of interactions between enhancer and *TFF3* did not explain the gain of *TFF3* expression at 3h post ligand stimulation.

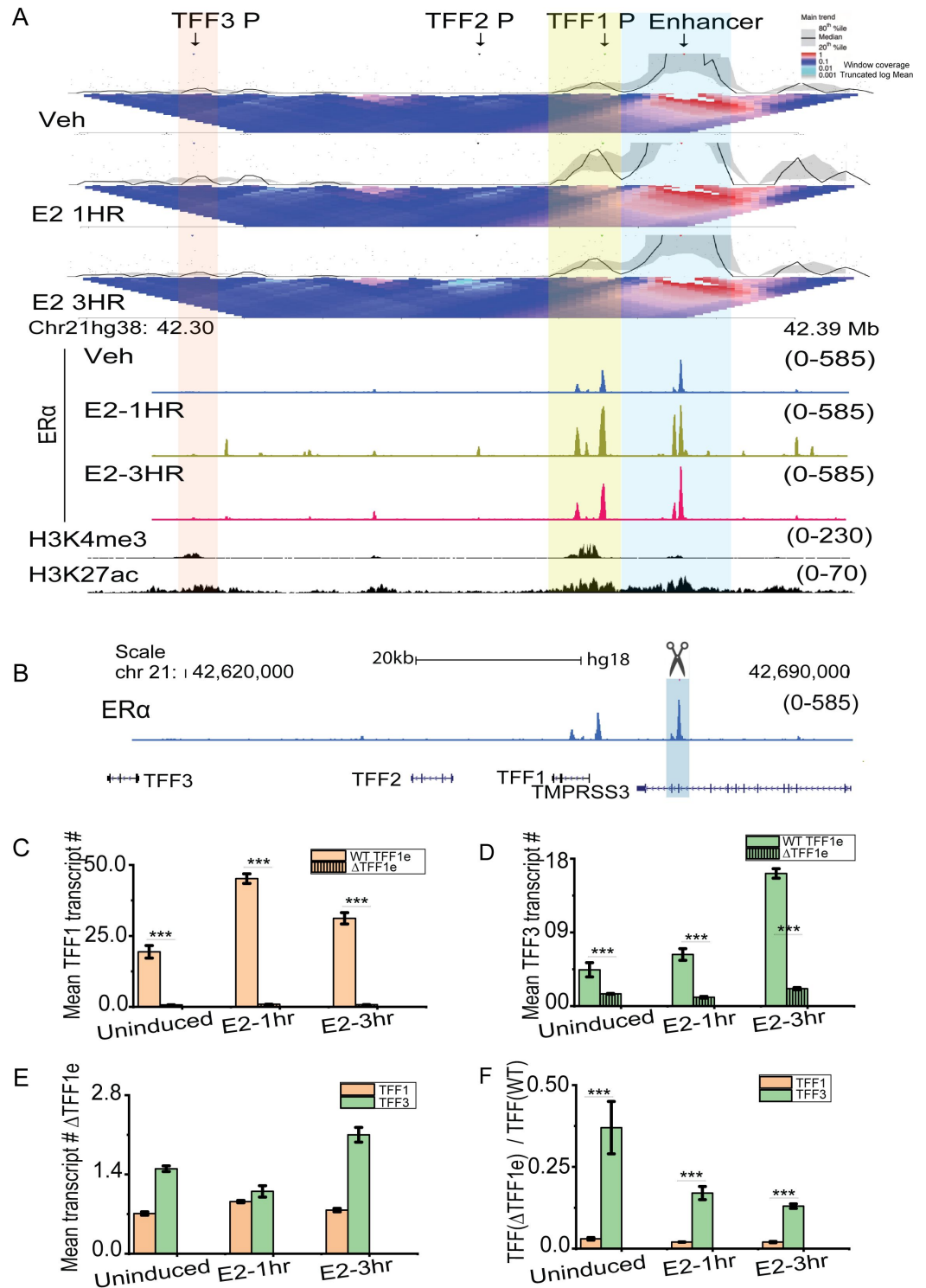


Figure 3.

Enhancer looping does not account for the differential expression of TFF1 and TFF3 genes

A. 4C-seq plot at TFF1 enhancer viewpoint, the interaction with the promoter is highlighted in yellow. The plot is overlaid with H3K27ac, ERα ChIP signal, and gene annotations.

B. Genome browser snapshot of *TFF1* region depicting ER α binding in WT lines. The first, second and third ER α ChIP-seq tracks are from WT cells that are vehicle-treated, E2-1h, and E2-3h, respectively. Blue highlighted regions represent the Δ *TFF1e* region.

C. The mean RNA numbers for In*TFF1* in WT (unshaded) and Δ *TFF1e* (shaded) MCF7 cells are depicted. The mean of means are shown, and error bars denote SEM from three repeats ($n > 650$, $N = 3$ for each WT and delete line). p-values were calculated by Student's two-tailed unpaired t-test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

D. The mean RNA numbers for In*TFF3* in WT (unshaded) and Δ *TFF1e* (shaded) MCF7 cells are depicted. The mean of means are shown, and error bars denote SEM from three repeats ($n > 650$, $N = 3$ for each WT and delete line). P-values were calculated by Student's two-tailed unpaired t-test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

E. The mean RNA numbers for In*TFF1* and In*TFF3* in Δ *TFF1e* MCF7 cells are depicted. The mean of means are shown, and error bars denote SEM from three repeats ($n > 880$, $N = 3$). p-values were calculated by Student's two-tailed unpaired t-test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

F. Ratio of In*TFF* in WT MCF7 to In*TFF* in Δ *TFF1e* MCF7 are depicted. The ratio was obtained by dividing the absolute RNA counts of the WT line by delete lines performed on different days but in the same order (replicate one of WT divided by replicate one of Δ *TFF1e*). The mean of means are shown, and error bars denote SEM from three repeats ($n > 650$, $N = 3$ for each WT and delete line). p-values were calculated by Student's two-tailed unpaired t-test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

This could mean that enhancer interaction is critical for *TFF1* expression, but is less important for *TFF3* expression. Thus, *TFF1* expression should be affected more severely upon deletion of the enhancer than *TFF3*. To test this, we investigated the expression of *TFF1* and *TFF3* in MCF-7 cells where the enhancer downstream to *TFF1* was homozygously deleted using CRISPR-Cas9 (referred to as the Δ *TFF1e* from hereon) (Fig. 3B) (Saravanan et al., 2021). Using smFISH, we quantified the intronic transcripts for *TFF1* and *TFF3* in the Δ *TFF1e* cells, compared to WT cells. We observed that the mean number of *TFF1* transcripts was reduced drastically in the Δ *TFF1e* compared to the WT (Fig. 3C). The reduction was far more substantial (49.23 ± 2.6 at 1hr and 40.75 ± 7.8 at 3h) for *TFF1* than *TFF3* (6.03 ± 1.8 at 1hr and 7.6 ± 0.7 at 3h) (Fig. 3D). The absolute transcript counts for *TFF1* and *TFF3* in the Δ *TFF1e* cells have also been shown for clearer visualization (Fig. 3E), as these are obscured when compared to WT. To get a sense of fold changes, we took a ratio of mean transcript counts in WT cells to Δ *TFF1e* cells at each time point. We observed that the drop in gene transcripts between WT and Δ *TFF1e* were several folds higher for *TFF1* compared to *TFF3* (Fig. 3F). This suggests that enhancer deletion has a more robust impact on the transcription of *TFF1* compared to *TFF3*. Nonetheless, *TFF3* was also affected even though, it does not loop with the enhancer. This is in accordance with the 4C-seq data where we observed prominent looping between the enhancer and *TFF1* but less so with *TFF3*. These results suggest that the enhancer plays a more important role in the expression of the primary gene while it has less impact on a gene located more distally but within the same TAD during course of signalling.

Levels of ER α in the nucleus dictate the extent of *TFF1* and *TFF3* inductions

After ruling out enhancer looping as the determinant of differential expression, we looked for other candidates that could regulate the differential gene expression. As discussed above, globally, E2-mediated gene expression is known to peak at 1h after stimulation and drop significantly by 3h (Hah et al., 2011). Upon ligand stimulation, ER α translocate into the nucleus, increasing mean

ERα intensity in the nucleus which is high at 1h and then decreases significantly by 3h due to degradation. These changes in intensities have been captured using immunofluorescence for ERα (Saravanan et al., 2020 [DOI](#)). We tested if the intensities of ERα in individual cells correlate with the expression of *TFF1* and *TFF3* at 1h and 3h of E2 signalling. Towards this, we combined smFISH with immunofluorescence for ERα. To improve contrast for the ERα signal, we additionally performed a chromatin retention assay to get rid of any chromatin unbound ERα (**Fig. 4A** [DOI](#)). The representative images show that the cells with very high levels of nuclear ERα (blue circle) exhibited low counts of both *TFF1* and *TFF3*. In contrast, the cells with medium levels of ERα (red circle) possessed higher *TFF1* than *TFF3*. Similarly, the cells with the lower levels of ERα (grey circle) showed higher *TFF3* expression as compared to *TFF1* (**Fig. 4A** [DOI](#)). Histograms depicting the ERα mean intensities across individual cells, showed that the nuclear level of ERα increases post 1h of induction and then goes down at 3h (**Fig. 4B, C** [DOI](#)), similar to *TFF1* expression (**Fig. 4D** [DOI](#)).

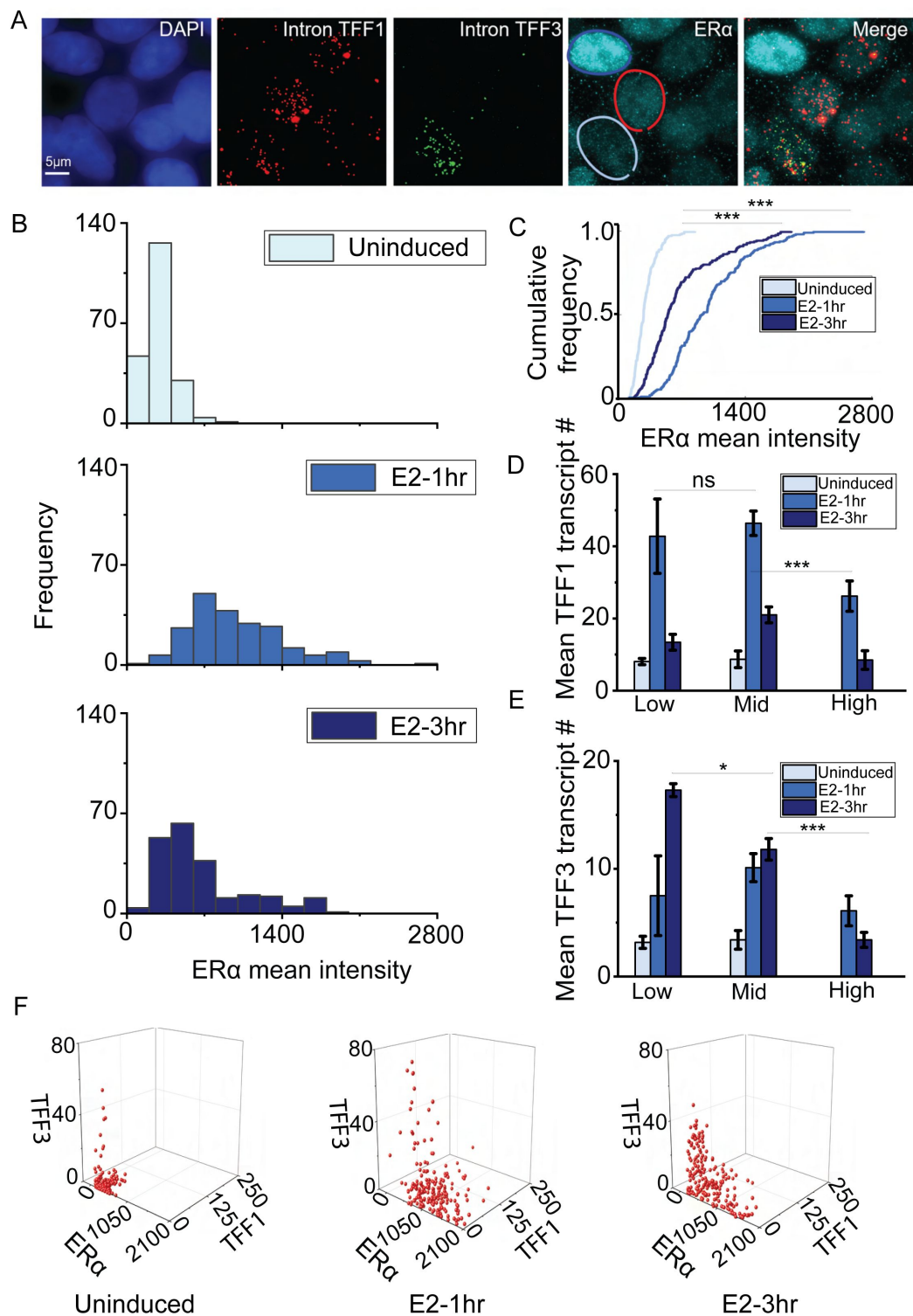


Figure 4.

Nuclear levels of the transcription factor ERα dictate the extent of expression of TFF1 and TFF3 genes

A. Representative images showing smFISH for InTFF1 and InTFF3 in combination with immunofluorescence for ERα (along with chromatin retention assay). The red circle denotes a cell with high ERα and low TFF1 and TFF3, the blue circle denotes a cell with medium ERα, high TFF1, and low TFF3 while the green circle denotes a cell with low ERα and high TFF3. The scale bar is 5 microns.

B. Histogram representing the distribution of ERα mean intensities in cells under uninduced, E2-1hr, and E2-3hr conditions (n=210). Intensities at 1hr are the highest while they shift to the left at 3hr and are lowest in uninduced cells. This is plotted from one experimental repeat out of three repeats as ERα intensities will vary from one immunofluorescence experiment to another.

C. Cumulative histogram representing the distribution of ERα mean intensities in cells under uninduced, E2-1hr, and E2-3hr conditions. p-values were calculated by the Mann-Whitney test and the significance is represented as : *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

D. ERα intensities were sorted into three categories namely low (intensities between 0-450 A.U.) mid (intensities between 450-1200 A.U.) and high (intensities between 1200-2100 A.U.). The mean and SEM of transcript count for InTFF1 in the three categories under uninduced, E2-1hr, and E2-3hr was plotted. Low and mid categories show the highest TFF1 mean. p-values were calculated by the Mann-Whitney test and the significance is represented as : *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$. This is plotted from one experimental repeat out of three repeats as ERα intensities will vary from one immunofluorescence experiment to another.

E. ERα intensities were sorted into three categories namely low (intensities between 0-450 A.U.) mid (intensities between 450-1200 A.U.) and high (intensities between 1200-2100 A.U.). The mean and SEM of transcript count for InTFF3 in the three categories under uninduced, E2-1hr, and E2-3hr was plotted. Low category shows the highest TFF3 mean. p-values were calculated by the Mann-Whitney test and the significance is represented as : *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$. This is plotted from one experimental repeat out of three repeats as ERα intensities will vary from one immunofluorescence experiment to another.

F. 3D plot representing the distribution of ERα, InTFF1, and InTFF3 on a cell-by-cell basis shows that cells with lower levels of ERα show higher counts for InTFF3. This is plotted from one experimental repeat out of three repeats as ERα intensities will vary from one immunofluorescence experiment to another.

To further corroborate this, we parsed the ERα population cells into 3 categories, namely low, medium, and high within the cells imaged at 1 and 3h. We plotted the mean counts of *TFF1* in each of these bins at different time points and observed that the mean count was higher in the mid-category (Fig. 4D). While for *TFF3*, the mean count was significantly higher in the low bin (Fig. 4E). The transcript counts for *TFF1* and *TFF3* against ERα intensities on a cell-by-cell basis (Fig. 4F), also showed this feature where very high levels of ERα in the nucleus were not conducive to the expression of either gene (Fig. 4F). To test if ERα had a causal role in intensity based expression of *TFF1* and *TFF3*, we increased the levels of ERα by overexpression of ERα-GFP (Fig. S5A). The representative images from the smFISH experiment in cells overexpressing ERα-GFP confirmed that *TFF1* and *TFF3* were downregulated in transfected cells while these genes were not perturbed in non-transfected cells in the neighbourhood (Fig. S5A,C). Meanwhile, the cells overexpressing ERα-GFP do not show any impairment in the expression of GAPDH (housekeeping gene) (Fig. S5A,B). As another control, we transfected the cells with EGFP-C1 (same backbone as ERα-GFP construct) and observed no effect on *TFF1* or *TFF3* (Fig. S5D,E). The data suggest that loss of *TFF1* and *TFF3* expression upon ERα overexpression was not a general effect of transfection stress but rather specific to ERα overexpression. These results indicate that high nuclear levels of ERα can be detrimental to the expression of genes it regulates. This could be due to the widespread

condensate formation, which in turn could sequester the transcriptional protein complexes and competitively abrogate transcription across multiple loci. Thus, the global level of ER α in the nucleus can predict the transcriptional status of specific genes. The binding of ER α at 1 and 3h is proportional to its nuclear levels (**Fig. 3A**), suggesting, its overexpression would lead to more binding in the genome, which is detrimental to gene expression.

LLPS perturbation down-regulates

TFF1 but supports *TFF3* expression

The data obtained from combined smFISH and ER α immunofluorescence indicates that ER α could be a determining factor in controlling the differential gene expression of *TFF1* and *TFF3*. Existing literature indicates that ER α -mediated condensate plays a role in E2-induced gene expression (Boija et al., 2018; Nair et al., 2019; Sabari et al., 2018; Saravanan et al., 2020). It led us to hypothesize that ER α -mediated condensate at the *TFF1* locus could be sequestering all the factors required for active transcription thus preventing activation of the *TFF3* locus at the active phase of E2 signalling. To validate this, we treated the cells with 3% 1,6-Hexanediol for 5 minutes, which is known to disrupt LLPS (Gamliel et al., 2022; Kroschwald et al., 2017). Following the treatment, we performed smFISH to look at the transcription of *TFF1* and *TFF3* in the same cell and observed a dramatic reduction in *TFF1* transcripts, whereas a statistically significant increase in *TFF3* was noted (**Fig. 5A**). Since ER α forms condensate only after estrogen stimulation, 1,6-Hexanediol had no effect on *TFF1* in the absence of E2 signalling (**Fig. S6**). Together, these results suggest that the functional loss of *TFF1* promoter transcription due to the dissolution of ER α condensate allowed the *TFF3* promoter in the neighbourhood to gain access to transcriptional machinery, leading to its upregulation. In order to test the generality of this observation, beyond the *TFF1* and *TFF3* locus, we divided E2-responsive genes into three categories, low, moderate, and high, based on their expression upon E2-40m stimulation. We observed a significant increase at 40m and down-regulation at 160m when the binding of ER α is reduced in the genome (**Fig. 5B**). Next, we tested the expression of their nearby upregulated genes at 160m compared to 40m post E2 stimulation. The nearby genes are within a distance of 1Mb from the E2-responsive genes, which is approximately the average size of a TAD. Indeed, we observed a significant upregulation of nearby genes at 160m when the expression of highly induced genes dropped. Additionally, the expression of these neighbouring genes was reduced at the peak of signalling (**Fig. 5B**), showing that at 40m of signalling, the acute activation of primary genes and sequestration of transcription machinery by these genes leads to the loss of expression of nearby genes.

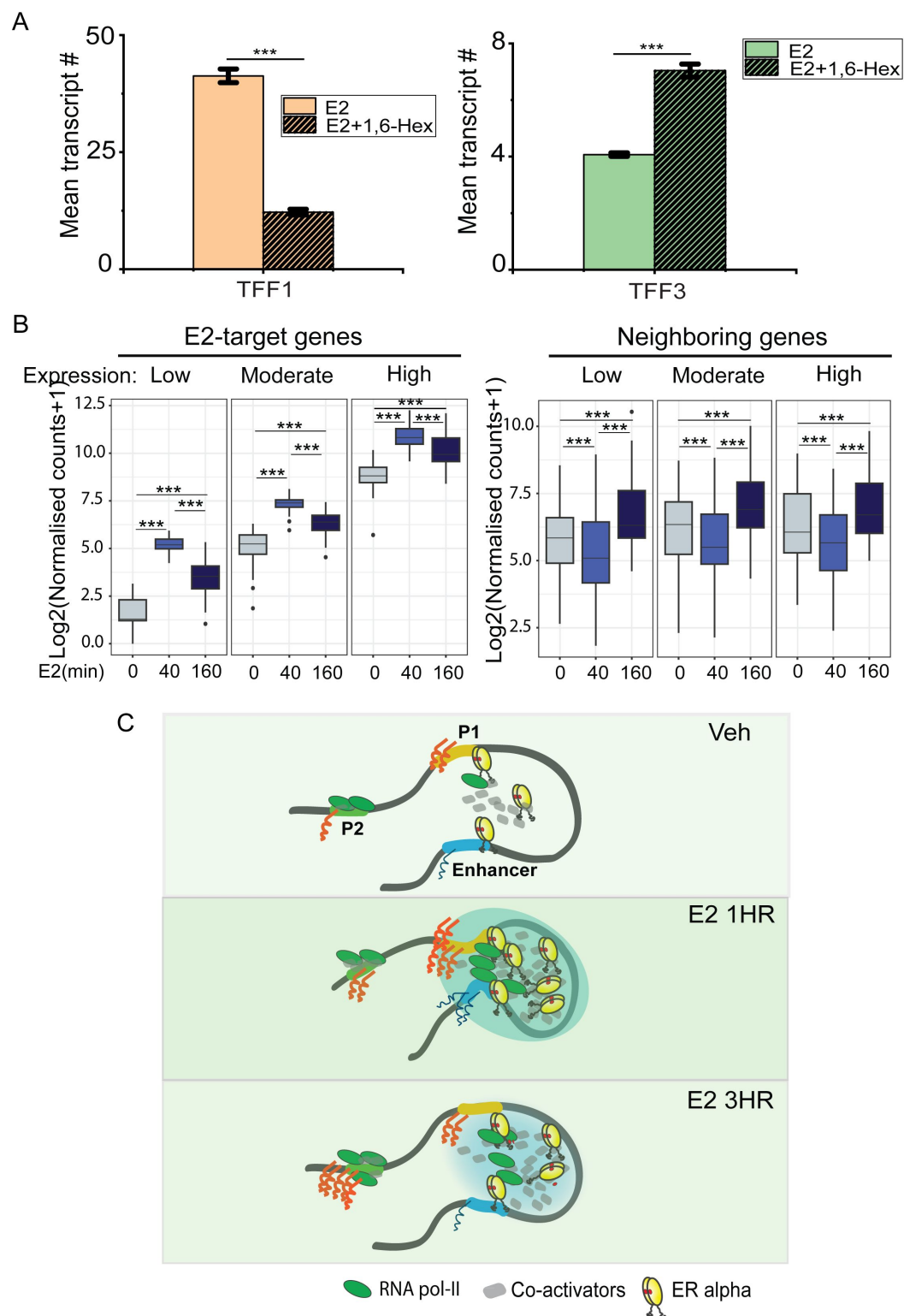


Figure 5.

ER α mediated phase separation could govern the differential expression of TFF1 and TFF3 genes

A. Mean transcript counts for InTFF1 and InTFF3 in control, and 3% 1,6-Hexanediol treated cells post 30 minutes of E2-induction. The mean of means are shown, and the error bars denote SEM from three repeats ($n > 880$, $N = 3$). p-values were calculated by Student's two-tailed unpaired t-test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

B. Boxplots showing DESeq2 normalized counts for low expressing, moderately expressing, and highly-expressing genes in the vehicle, E2-40m, and E2-160m respectively (left). Boxplots showing DESeq2 normalized counts for genes near low expressing, moderately expressing, and highly expressing genes in the vehicle, E2-40m, and E2-160m, respectively (right). The p-values in the boxplots were calculated using the Wilcoxon rank-sum test. The boxplots depict the minimum, first quartile, median, third quartile, and maximum values, along with outliers.

C. Model depicting the signaling under uninduced, E2-1hr, and E2-3h conditions-1. Activation of TFF1 locus by ligand induction. 2. During the active phase, liganded ER α binds on enhancer and promoter. Together these in 3D interact, manifesting as ER α puncta, which results in robust expression of target genes, 3. The phase-separated condensate leads to the sequestration of transcriptional machinery. 4. This favours intra-TAD enhancer: promoter interactions to facilitate robust gene transcription. Upon ER α degradation at 3h, these condensates disappear, and transcriptional machinery becomes available to other promoters leading to their somewhat increased gene transcription.

Discussion

smFISH allowed us to simultaneously capture allele level transcription of two genes, *TFF1* and *TFF3*, located in the same TAD, at single cell level during the peak and fall of signalling. We were able to capture the anti-correlated expression of these two genes, revealing intricate regulatory feedback between acutely activated enhancer-dependent gene, *TFF1*, which caused the dysregulation of the non-enhancer targeted gene, *TFF3*, within the same TAD (**Fig. 5B** [↗](#)).

Our data suggests that while condensate formation on enhancer allows robust activation of the target gene, however, it negatively impacts the expression of other neighbouring genes potentially due to the sequestration of transcriptional machinery from these genes. When the condensates dissolve, the locally enriched transcriptional machinery is available to the other loci, potentially allowing for increased transcription of these genes, that are not the direct target of enhancers (**Fig. 5C** [↗](#)). Forced dissolution of condensates allows the expression of non-enhancer target genes in the cells, although at low levels. The data also suggests that, globally, the genes next to ligand-induced genes are upregulated only at the fall of signalling as an indirect consequence of excess polymerase availability in the neighbourhood.

Non-enhancer target genes are also regulated indirectly in the same TAD

ER α binding strength increases at the *TFF1* enhancer at 1h of ligand stimulation ([Saravanan et al., 2020](#) [↗](#)). Such estrogen-induced binding of ER α leads to robust ligand-induced activation of *TFF1* gene. While *TFF3* promoter remains unbound by ER α (**Fig. 3A** [↗](#)) and does not interact with the *TFF1* enhancer throughout the course of signalling. These data suggest that *TFF3* expression is both ER α and enhancer independent. Despite non-dependence, *TFF3* expression was mutually exclusive to *TFF1* expression and was negatively impacted in the absence of the enhancer. The latter could be due to the enhancer-mediated repositioning of the entire TAD that benefitted the expression of *TFF3* and not due to the direct interaction between the enhancer and the *TFF3* gene.

The mean expression of *TFF1* was manyfold higher than *TFF3*, thus explaining its dependence on an active upstream enhancer. The enhancer of *TFF1* interacts with the gene at the peak of signalling and shows some loss of interaction at 3h when signalling recedes. As stated above, the enhancer does not interact with *TFF3*, however, the enhancer forms loop with *TFF3* upon deletion of the *TFF1* promoter (Oh et al., 2021 [↗](#)). The ERα condensates are formed at the *TFF1* locus upon signalling (Saravanan et al., 2020 [↗](#)), which is reduced upon enhancer deletion, suggesting its causal role in condensate formation. Corroborating with this, the enhancer of *TFF1* engages in the recruitment of megadalton complex (Liu et al., 2014 [↗](#)) in MCF-7 cells, which could facilitate phase-separation (Nair et al., 2019 [↗](#), Saravanan et al., 2020 [↗](#)). The cells that expressed *TFF1* did not express *TFF3* as efficiently, suggesting Pol2 was potentially sequestered from the *TFF3* promoter. However, at 3h when liganded ERα degrades leading to dissolution of condensates on enhancer, *TFF3* expression increased, suggesting a local redistribution of active transcription machinery to other genes within the same chromatin domain. Even at 1h, when condensates were perturbed by 1,6 HD, the expression of *TFF3* increased indicating, the access to Pol2 pool by *TFF3* promoter. The data explains the negative correlation between *TFF1* and *TFF3* expression at the peak of signalling and robust activation of *TFF3* at 3h when *TFF1* expression decreased at single cell level. The presence of significant numbers of *TFF1* nascent RNA in the nucleus in our data could also be due to a time lapse between transcription and splicing, as seen for the genes that are early responders during signaling (Zambrano et al., 2020 [↗](#)).

High and low levels of ERα were detrimental to enhancer mediated activation of TFF1

We observe that *TFF1* expression is the highest in cells with medium levels of ERα, while it shows low expression in cells with low or high levels of ERα (Fig. 4D, F [↗](#)). This points to an inverted U-shaped response of *TFF1* to ERα, which is a nonmonotonic response to estrogen, as has been shown (Lebedeva et al., 2012 [↗](#); Vandenberg et al., 2012 [↗](#)). A nonmonotonic response can be described as an inverted U-shaped curve with maximal response occurring in the middle and reducing substantially at very low or very high doses (Vandenberg et al., 2012 [↗](#)). Further, we observe that for *TFF3*, the maximal expression occurs at low levels of ERα (Fig. 4E, F [↗](#)). This could be indicative of a response such that a low concentration is enough to meet the threshold for expression beyond which all other higher doses attenuate the expression (Lebedeva et al., 2012 [↗](#); Vandenberg et al., 2012 [↗](#)). This could also suggest that ERα levels have a varied effect on different genes and their expression depending on the extent of reliance of the gene on ERα for its expression. This phenomenon, called the prozone effect, is characterized by the inhibition of multimeric complexes when one of the constituents of the complex are present at very high concentrations as has been shown for *TFF1* (Lebedeva et al., 2012 [↗](#)). In the context of condensates on *TFF1* gene, we speculate that very low levels of TF may not reach or exceed the critical concentration required for LLPS (for *TFF1*), but this poor transcriptional state of *TFF1* could be favourable for *TFF3*. Alternatively, a very high concentration of the TF could result in the formation of highly dense homogenous condensates, effectively reducing transcription in a situation akin to the prozone effect (Chong et al., 2022 [↗](#); Lebedeva et al., 2012 [↗](#); Ryu et al., 2024 [↗](#)).

Interestingly, the levels of ERα do not determine the expression of other E2 target genes like *GREB1* and *MYC* (Stossi et al., 2020 [↗](#)); however, we observed that overexpression of ERα causes downregulation of *TFF1* (Fig. 5SA [↗](#)) and this is in agreement with what has been observed for the expression status of *TFF1* in cell line with high ERα compared to cell lines with low ERα (Lebedeva et al., 2012 [↗](#)).

ER is known to be expressed in nearly 70% of breast cancers, and many of these are treated using estrogen receptor modulators and degraders (Arnesen et al., 2021 [↗](#)). Therefore, understanding the effect of ERα on the expression of estrogen-responsive genes depending on whether they are

target genes or non-target genes is of therapeutic importance as several non-target genes have been shown to be differentially regulated in ER-mutant breast cancer cells (Arnesen et al., 2021 [↗](#)).

Together, our study underscores the indirect effects of ligand-induced chronic gene transcription that is dependent on enhancer activation and phase separation within a TAD.

Materials and Methods

Cell culture

MCF-7 cells (WT scramble control and $\Delta TFF1e$ were generated in Saravanan et al., 2020 [↗](#)) cells were cultured in non-stripping media consisting of DMEM (Gibco, 12100-046) supplemented with 10% FBS (Gibco, 16000-044) and 1% Penicillin-Streptomycin-Glutamine (Gibco, 10378-016) in a 5% CO₂ humidified incubator at 37 °C (unstripped MCF-7). Cells were passaged and seeded into glass bottom dishes in non-stripping media for 24h and allowed to reach 60–80% confluency. These cells were then hormone-stripped for three days in stripping media containing phenol red-free DMEM (Gibco, 21063-029) supplemented with 5% charcoal-stripped FBS(Gibco,12676029) and maintained in the humidified incubator at 37 °C (stripped MCF-7).

In order to induce the estrogen transcriptional responses, on the third day cells were treated with β -estradiol (E2758, Sigma-Aldrich) at 100nM concentration for various periods as mentioned in the respective Figures. For untreated control, cells were either treated with equal microliters of ethanol on the third day or with ER α inhibitor ICI182780 (1047, Tocris Biosciences) at 100nM concentration for 24 h after two days of stripping.

For perturbation of LLPS in E2-induced condition, cells were incubated with media containing 3% 1,6-Hexanediol for 5 minutes (after 30 minutes of E2 treatment) followed by removal of 1,6-Hexanediol and recovery for 25 mins in E2 containing media.

For experiments with overexpression of EGFP-ER α or EGFP, cells were transfected 24 hours before E2 induction using X-tremeGENE™ HP DNA Transfection Reagent (XTGHP-RO) in stripping media.

Probe for smFISH

Probes were designed, custom-made, and tagged with indicated fluorophores (Quasar 570 or Cal 610) by Biosearch Technologies (<https://www.biosearchtech.com/products/rna-fish> [↗](#)). Probes were made to target the intronic or exonic regions of the *TFF1* and *TFF3* genes. Probe sequences can be found in **supplementary table 1** [↗](#) [↗](#) [↗](#) in supplementary materials.

Chromatin retention assay and smFISH

For smFISH experiments without chromatin retention assay cells, the cells were treated with estradiol or vehicular control for indicated times. Following this, the cells were washed with nuclease-free 1X PBS (Ambion, AM9624) twice. This was followed by fixation using 4% paraformaldehyde (PFA, Sigma, P6148) in 1X PBS for 10 minutes at room temperature. The fixative was removed and cells were washed twice with 1X PBS. The cells were permeabilized using 70% ethanol at 4°C overnight. On the next day, the permeabilizing agent was removed and the cells were washed twice with 1X PBS. The cells were washed once with a wash buffer (10% Formamide (Ambion, 9342) and 2X SSC (Ambion, AM9763) in nuclease-free water) for 5 minutes at room temperature. The wash buffer was aspirated and the cells were incubated with a hybridization mix (100 μ l of mix contains 10% formamide, 89 μ l hybridization buffer, and 1 μ l each of indicated probes) overnight at 37°C in a humid chamber. The next day, the solution was removed and cells were washed with the wash buffer at 37°C for 30 minutes. To stain the nuclei, DAPI (Invitrogen, D1306; 2 μ g/ml) in wash buffer was added to the cells and incubated for 10 minutes at 37°C. The

cells were then washed with 2X SSC for 5 minutes at 4°C. The solution was removed and the cells were covered with a few drops of the mountant Vectashield (Vector Labs). The plates were imaged after at least 1h.

For smFISH experiments with chromatin retention assay cells were treated with estradiol or vehicular control for indicated times. Following this, the cells were treated with CSK buffer (consisting of 10 mM PIPES buffer, 100 mM NaCl, 3 mM MgCl₂, 300 mM sucrose, and 0.7% Triton-X 100) for 15 minutes at room temperature. After this, the cells were washed with nuclease-free 1X PBS twice. This was followed by fixation using 4% paraformaldehyde in 1X PBS for 10 minutes at room temperature. The fixative was removed and cells were washed twice with 1X PBS. The cells were permeabilized using 0.3% Triton-X 100 (Sigma, T8787) in 1X PBS for 10 minutes at room temperature. The permeabilizing agent was aspirated and the cells were washed twice with 1X PBS. This was followed by a washing step using the wash buffer for 5 minutes at room temperature. Following this, the cells were incubated with the hybridization mix to which the antibody of interest was also added at appropriate dilution. The next day, the hybridization mix containing probes and antibodies was removed. Cells were washed twice with 1X PBS. Then the cells were incubated with 1X PBS containing a secondary antibody against the antibody of interest at room temperature for 2h. This was followed by incubation with the wash buffer at 37°C for 30 minutes. To stain the nuclei, DAPI in wash buffer was added to the cells and incubated for 10 minutes at 37°C. The cells were then washed with 2X SSC for 5 minutes at 4°C. The solution was removed and the cells were covered with a few drops of the mountant Vectashield. The plates were imaged after at least 1h.

Antibody staining/ immunofluorescence

The primary antibody against ER α (Santa Cruz, sc8002(F10)) was added to the hybridization mix at the dilution of 1:400 and incubated overnight at 37°C. The next day, an anti-mouse Alexa Fluor™ 488 secondary antibody (Invitrogen, A11029) was added to 1X PBS at the dilution of 1:1000. Cells were incubated for 2h followed by the continuation of the smFISH protocol as indicated above.

Image acquisition

The plates were imaged on an Olympus IX83 inverted widefield fluorescence microscope with a Retiga 6000 CCD monochrome camera (QImaging). The images were acquired using a 60X, 1.42 N.A. oil immersion objective or a 100X, 1.4 N.A. oil immersion objective. The z-step size was 0.3 μ m and 35 slices were acquired. The resolution at which the images were acquired is 2752 x 2208. Narrow band-pass filters were used to distinguish the signal from Quasar 570 and Cal 610 labeled probes (ChromaTechnology-49309 and 49310).

Image analysis and representation

The images of mRNA channels were subjected to rolling ball background subtraction across the entire Z-stack using Fiji. For representative images, the stacks were Z-projected in Fiji. Transcripts were counted using either Rajlabimagetools (courtesy of Arjun Raj lab) or an in-house MATLAB (Mathworks) script based on earlier work (Femino et al., 1998 [DOI](#); Raj et al., 2008 [DOI](#)). Briefly, the code uses a nuclear mask to count the intron-containing RNA present in the nucleus (InTFF), and another mask for the whole cell to count the mature mRNA, detected by exonic probes (ExTFF), that are present in the nucleus as well as the cytoplasm. The mean intensity for the ER α immunofluorescence channel was also quantified using the MATLAB script on a cell-by-cell basis using the nuclear mask.

Graphing and Statistics

The graphs were plotted using Python 3, MATLAB, and Origin Pro and edited using Adobe Photoshop. To perform a student's t-test for data combined from three repeats, GraphPad (<https://www.graphpad.com/quickcalcs/ttest1/> [DOI](#)) was used. Non-parametric tests were performed in case of

single cell data. For this Mann-Whitney test was used. Significance is represented as : *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$. The specific test used has been mentioned in the relevant figure legends.

Circular Chromatin Conformation Capture-seq

4C was performed as per the protocol described in [van de Werken et al., 2012](#) with minor variations. MCF7 cells were fixed with fresh formaldehyde (1.5%) and quenched with glycine (125mM) followed by washes with ice-cold 1XPBS (2X) and scraped, pelleted, and stored at -80°C . Lysis buffer [Tris-Cl pH 8.0 (10mM), NaCl (10mM), NP-40 (0.2%), PIC (1X)] was added to the pellets and homogenized by Dounce homogenizer (20 stroked with pestle A followed by pestle B). The 3C digestion was performed with DpnII (200 units, NEB) and ligation was performed by T4 DNA ligase and ligation mix [Triton X-100 (1%), 1× Ligation buffer (10X Ligation buffer-Tris-Cl pH 7.5 (500mM), MgCl₂ (100mM), DTT (100mM), BSA (0.105mg/ml), ATP (1.05mM)]. The ligated samples were purified by PCI and subjected to ethanol precipitation. The pellet was eluted in 1X TE (pH 8.0) to obtain the 3C library. The 4C digestion was performed by NlaIII (50 units, NEB), and the samples were ligated, purified, and precipitated similar to the 3C library to obtain the 4C library. The 4C library was subjected to RNaseA treatment and purified with the QIAquick PCR purification kit. The concentration of the library was then measured by Nanodrop and subjected to PCRs using the oligos for the enhancer viewpoint. The samples were next PCR purified using the same kit and subjected to next-generation sequencing with Illumina HiSeq2500/ NOVA seq. The 4C oligos are listed in [supplementary table 5](#).

4C Data Analysis

The sequenced reads in fastq file were demultiplexed by matching the appropriate primer sequences for each condition without allowing for any mismatches. Demultiplexed reads were processed using 4cseqpipe software. Restriction site tracks were created for the hg38 human genome by mentioning the restriction sites of the first cutter as GATC and that of the second cutter as CATG. The phred scores of demultiplexed reads were changed to phred64 format and the fastq files were converted into raw format. Further, valid 4C reads were mapped to the generated restriction site tracks. Unique fragment ends/non-unique fragment ends were used. The mapped reads were normalized and near cis domainograms at a maximum height of 0.1 were created by using the truncated log mean statistic with a trend resolution of 1kb for the genomic region chr11:42300000-42400000.

RNA Isolation, cDNA synthesis and PCR

Cells were lysed in 1 ml of Trizol (Thermo Fisher Inc.). 200 μl chloroform was added to the sample, briefly vortexed and centrifuged at 12K rpm for 12 mins. The aqueous phase was carefully collected and transferred to the fresh tube. 1 volume of isopropanol was added to the sample and incubated at room temperature for 10 mins to precipitate the RNA. The samples were centrifuged at 12K rpm for 12 mins, supernatant was discarded without disturbing the pellet. The pellet obtained was washed with 75% ethanol. Pellet was air dried and dissolved in RNase free water. RNA obtained was treated with ezDNase (Invitrogen) to remove the traces of contaminating DNA. 1 μg of RNA was used for each cDNA synthesis reaction by Superscript IV (Invitrogen) and random hexamers as per manufacturer's recommendation. The CFX96 touch (Biorad) real time PCR was used for qRT-PCRs. The fold changes were calculated by the $\Delta\Delta\text{Ct}$ method and individual expression data was normalized to GAPDH mRNA. The p -values were calculated by Student's unpaired two-tailed t -test from independent four biological replicates. qRT-PCR primers are listed in [Supplementary Table 6](#).

GRO-Seq analysis

Fastq files from GEO accession number GSE43836 were downloaded from European Nucleotide Archive. Reads with base quality <20 in a sliding window of 4 bases and with a length of <36 were removed using Trimmomatic 0.39. Trimmed reads were aligned using bowtie2 2.5.1 with default parameters. Duplicate reads from the alignment files were removed using samtools 1.16.1. De-duplicated aligned reads were assigned in a strand-specific manner to transcript feature of the hg38.ncbiRefSeq.gtf.gz file by allowing multi-mapping reads and considering the largest overlap in case of overlapping features using featureCounts v2.0.3. CPM normalised strand-specific signal files with a bin size of 1 bp were generated using bamCoverage 3.5.1. Differential gene expression analysis was performed with default parameters using Deseq2 1.36.0. Upregulated genes were defined as those with adjusted p-value <0.05 and log₂(FC) >1 in their respective conditions. Top, Middle and bottom 10% of upregulated genes based on their base mean value in E2-40m vs VEH condition were subsetting as highly expressing, moderately expressing and low expressing upregulated genes. The closest upregulated genes in E2-160m vs E2-40m near highly expressing, moderately expressing and low expressing upregulated genes in E2-40m vs VEH condition were identified using bedtools closest v2.30.0. Only genes within a genomic distance of 1Mb were considered. Log₂ transformed DEseq2 normalised counts with the addition of the arbitrary value 1 was used to compare the gene expression trends across time points and categories of upregulated genes. All plots were generated using R 4.2.2.

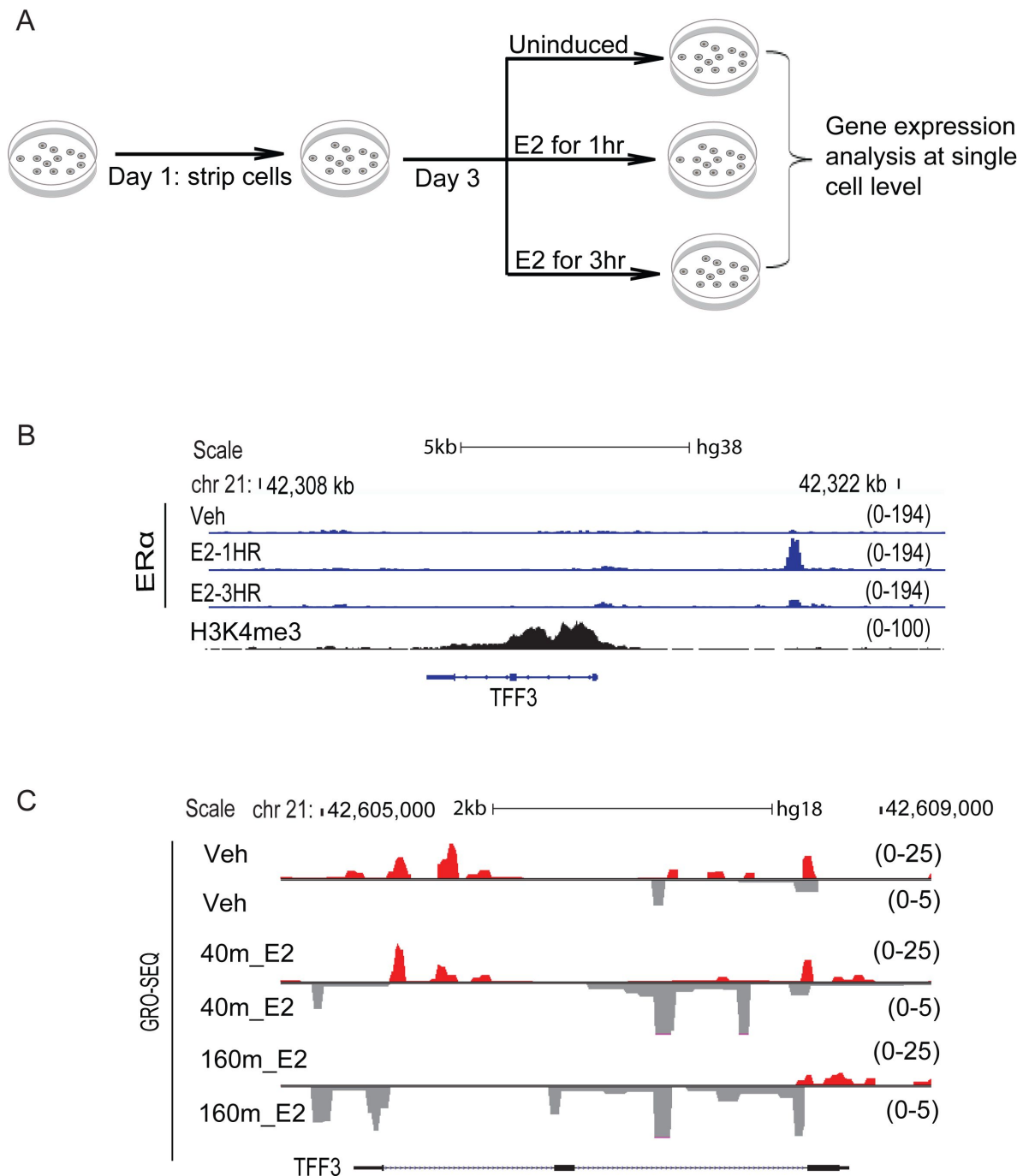
Additional information

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Contribution

The project was conceived by DN, AM, DB and ZI. Experiments were performed by DB and ZI. 4C and GRO-seq data analysis was performed by SN. The manuscript was written by DB, ZI, DN and AM. All authors approved the manuscript.



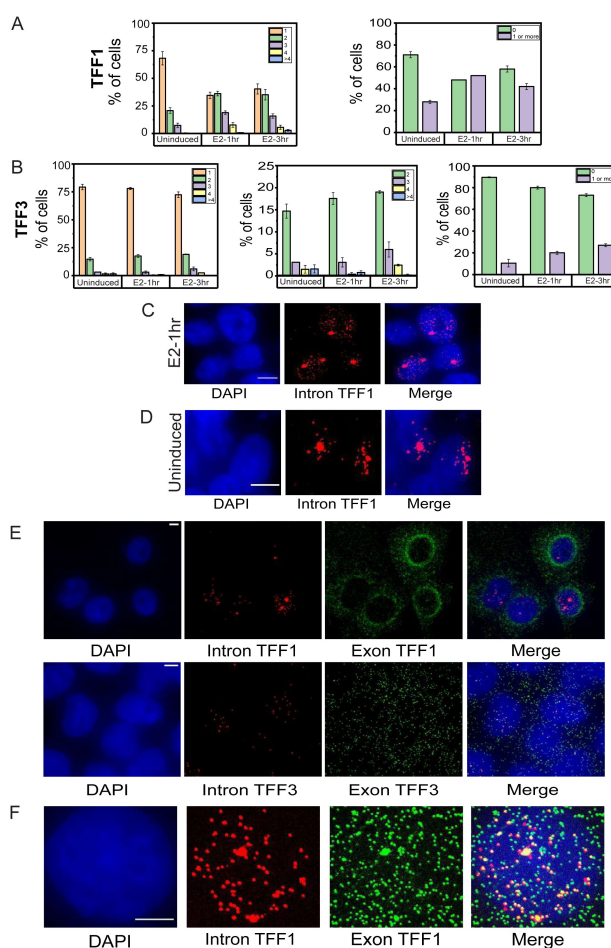
Supp. Figure 1.

Experimental design

A. Schematic depicting the experimental design. Cells were cultured in complete media for 24h followed by stripping for 3 days. Finally, cells were induced for different durations with E2/Vehicle followed by different assays like smFISH, 4C-seq, smFISH-IF, etc.

B. Zoomed in region around TFF3 gene, UCSC genome browser snapshots showing the binding of ERα and H3K4me3 signal. First, second and third ERα ChIP-seq tracks are in vehicle treated, E2 treated for 1h and E2 treated for 3h in WT cells respectively

C. UCSC genome browser snapshots showing Gro-seq signal for *TFF3* gene. First, second and third Gro-seq tracks are from vehicle-treated, E2-40m and E2-160m in WT cells, respectively.



Supp. Figure 2.

Dynamic induction and RNA localization of *TFF1* and *TFF3* transcription across cell populations using smRNA FISH

A. Bar graph depicting the percentage of cells with 1,2,3,4, or greater than 4 sites of transcription for TFF1 (left) is shown. The graph shows the mean of means from different repeats of the experiment, and error bars denote SEM ($n > 200$, $N = 3$). Only the cells with at least one allele firing were counted and cells with no alleles were not included in this. The graph on right shows the number of cells with zero or non-zero number of alleles firing.

B. Bar graph depicting the percentage of cells with 1,2,3,4 or greater than 4 sites of transcription for TFF3 (left) is shown. The graph shows the mean of means from different repeats of the experiment, and error bars denote SEM ($n > 200$, $N = 3$). Only the cells with at least one allele firing were counted and cells with no alleles were not included in this. The graph in the middle shows the number of cells with 2,3,4 or greater than 4 sites of transcription for TFF3. The graph on the right shows the number of cells with zero or non-zero number of alleles firing.

C. Images from single molecule RNA FISH experiment showing transcripts for InTFF1 in cells induced for 1 hour with E2. The image shows that when a single allele of TFF1 is firing, the transcripts show a more spatially restricted localisation. The scale bar is 5 microns.

D. Images from single molecule RNA FISH experiment showing transcripts for InTFF1 in uninduced cells. The image shows that when a single allele of TFF1 is firing and transcription is low, the transcripts show a more spatially restricted localisation. The scale bar is 5 microns.

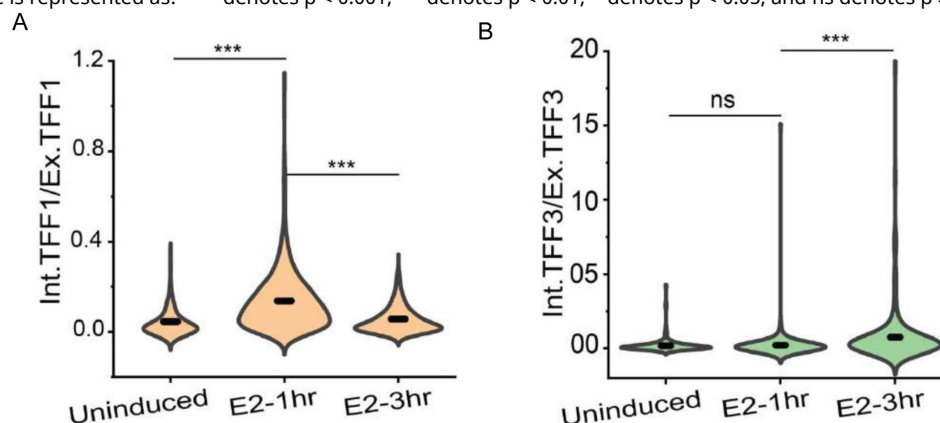
E. 60X Representative images from a single molecule RNA FISH experiment showing transcripts for InTFF1 and ExTFF1 (top) and InTFF3 and ExTFF3 (bottom). The image shows that there is no intronic signal in the cytoplasm, while exonic signals can be found both in the nucleus and the cytoplasm. The scale bar is 5 microns.

F. 60X Representative images from single molecule RNA FISH experiment showing transcripts for InTFF1 and ExTFF1. The image shows that all intronic signals are colocalized with exonic signals, but all exonic signals are expectedly not colocalized with intronic signals, representing more mature mRNA. The scale bar is 5 microns.

Supp. Figure 3.

TFF1 and *TFF3* expressions show opposite trends during the E2-signaling time-course

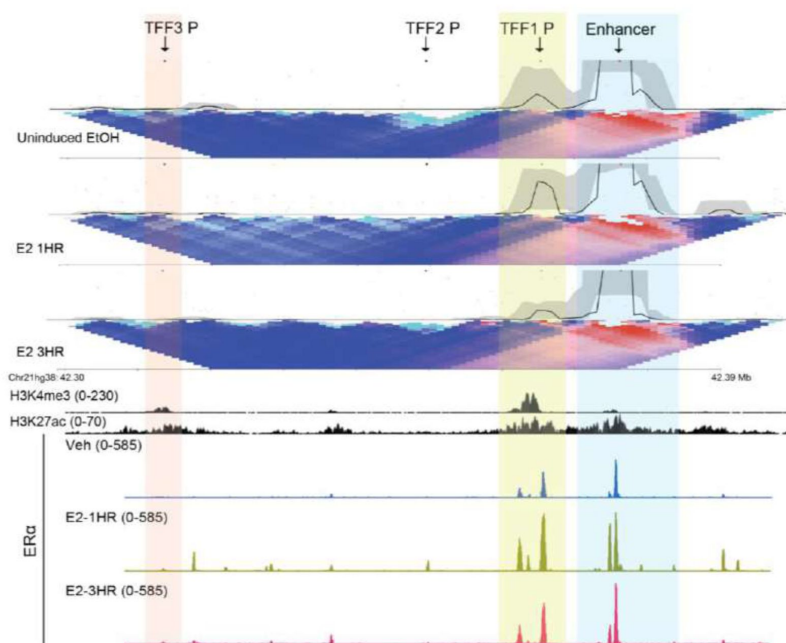
A. Violin plot showing the ratio of total intronic to absolute exonic *TFF1* counts are depicted. Absolute exonic counts are calculated by subtracting total intronic transcripts from total exonic transcripts. The graph shows the distribution of ratios combined from three different repeats. Error bars denote SEM. p-values were calculated by the Mann-Whitney test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, and ns denotes $p > 0.05$.
 B. Violin plot showing the ratio of total intronic to absolute exonic *TFF3* counts are depicted. Absolute exonic counts are calculated by subtracting total intronic transcripts from total exonic transcripts. The graph shows the distribution of ratios combined from three different repeats. Error bars denote SEM. p-values were calculated by the Mann-Whitney test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, and ns denotes $p > 0.05$.

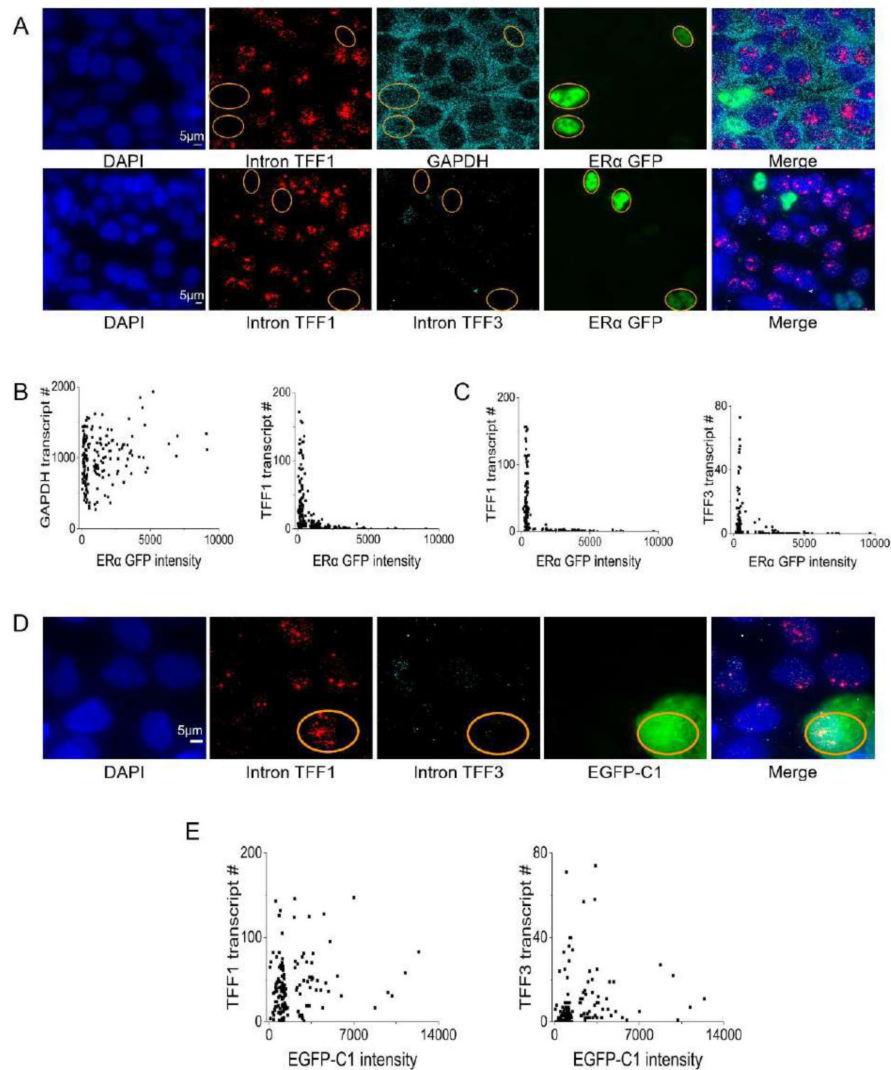


Supp. Figure 4.

TFF1 enhancer does not change target promoters during signalling time-course

4C-seq plot at *TFF1* enhancer viewpoint, the interaction with the promoter is highlighted in yellow. The plot is overlaid with H3K27ac, ERα ChIP signal, and gene annotations. There is no substantial interaction between the enhancer and *TFF3* locus at any time point, while the interaction between the enhancer and *TFF1* locus increases at 1hr and decreases at 3hr.





Supp. Figure 5.

Levels of ERα in the nucleus dictate the extent of *TFF1* and *TFF3* inductions

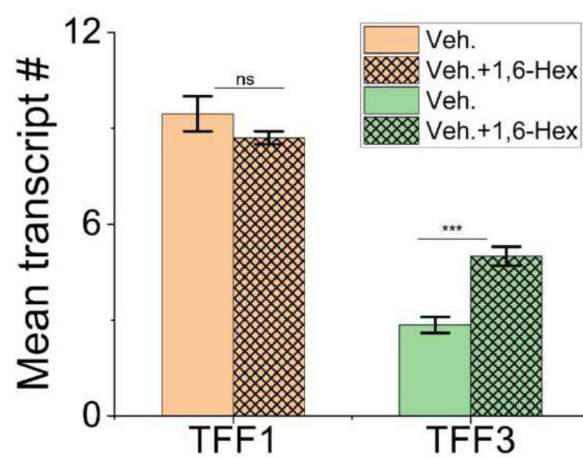
A. Representative images from smFISH experiments showing *InTFF1* and GAPDH (top panel) and *InTFF1* and *InTFF3* (bottom panel) in cells overexpressing ERα-GFP. Yellow circles show cells that are GFP positive, and the transcripts associated with them. Note the visibly fewer *TFF1*, and *TFF3* transcripts in the ERα-GFP positive cells, while GAPDH transcripts remain indistinguishable. This shows that ERα-GFP overexpression specifically affects the transcription of E2-regulated genes like *TFF1* and *TFF3* and not a housekeeping gene like GAPDH. This is further quantified in B and C.

B. Scatter plots showing the distribution of ERα-GFP intensities with GAPDH or *InTFF1* transcript counts on a cell-by-cell basis from the experiment in the first row of A. Cells with high ERα-GFP expression can have high GAPDH expression, but such cells necessarily have low *TFF1* counts.

C. Scatter plots showing the distribution of ERα GFP intensities with *InTFF1* or *InTFF3* counts on a cell-by-cell basis from the experiment in the second row of A. As in B, high ERα-GFP expression leads to low expression of E2-regulated genes like *TFF1* or *TFF3*.

D. Representative images showing smFISH for *InTFF1* and *InTFF3* in cells overexpressing EGFP-C1. Yellow circles show cells that are GFP positive, and the transcripts associated with them. Just expressing GFP in the same backbone as the ERα-GFP plasmid, does not downregulate *TFF1*, and *TFF3* expression; indicating that the downregulation effect is specific to ERα overexpression and not a generic effect of cell transfection. This is quantified in E.

E. Scatter plots showing the distribution of EGFP-C1 intensities with *InTFF1* or *InTFF3* transcript counts on a cell-by-cell basis. There is no obvious correlation between the levels of EGFP-C1 and *TFF1* or *TFF3*, and a cell with high EGFP levels may well have large *TFF1* or *TFF3* transcript counts.



Supp. Figure 6.

LLPS perturbation down-regulates *TFF1* but supports *TFF3* expression

Mean transcript counts for *InTFF1* and *InTFF3* in control, and 3% 1,6-Hexanediol treated cells in the absence of E2-induction show no difference in *TFF1* counts on 1,6-Hexanediol treatment. The bar denotes the mean of means from two repeats, and the error bars denote SEM. p-values were calculated by Student's two-tailed unpaired t-test, and the significance is represented as: *** denotes $p < 0.001$, ** denotes $p < 0.01$, * denotes $p < 0.05$, ns denotes $p > 0.05$.

Supp. table 1

IntFF3 probe sequence

Probe Sequence (5' to 3')	Probe sequence name
catcttctcttttcttgt	Int_TFF3_1
ctgaagtcagggttttctc	Int_TFF3_2
taaatgaaccttgtgcctgc	Int_TFF3_3
ggcaggggtgctatgcaatg	Int_TFF3_4
actgggttaatggagcagc	Int_TFF3_5
ataaaagccatgccctcatt	Int_TFF3_6
tttgacctgctcagaagac	Int_TFF3_7
cgtaattgcttatcctctg	Int_TFF3_8
tcttacattccttctcttg	Int_TFF3_9
ggaaaatggcaatcacgtgc	Int_TFF3_10
cagaagttcactgcgtttca	Int_TFF3_11
caagcctctatgtctgtatg	Int_TFF3_12
ccccaatatttcatgtttc	Int_TFF3_13
ctgaattgtgagcatgtcc	Int_TFF3_14
tgcttttctcttttctgc	Int_TFF3_15
cgccatagagatcattgcaa	Int_TFF3_16
aaaccgtgacaggagaggca	Int_TFF3_17
cttggccctttgttttaaat	Int_TFF3_18
ccttcattgttgaccaaca	Int_TFF3_19
ccctgaaccagaaacaaa	Int_TFF3_20
cttctcgggaagtcacatac	Int_TFF3_21
catttgccctaagtgctttc	Int_TFF3_22
gtctgttaaatgctcagtc	Int_TFF3_23
ccttaggaagatgcacttc	Int_TFF3_24
cgagttcaaccactgctgaa	Int_TFF3_25
agtgtgtttgcttactttg	Int_TFF3_26
agcccagaaaaggacaggag	Int_TFF3_27
ttggaacagggtgtgtgtg	Int_TFF3_28
taagtgtgaagcctctgctc	Int_TFF3_29
aggggacagaagaggacagc	Int_TFF3_30
tgcgatgccatcatttgaa	Int_TFF3_31
gttactaaatcagttccctc	Int_TFF3_32
gtttccacttctctcaagaa	Int_TFF3_33
ggccgatttgcaaaaacagc	Int_TFF3_34
ggggatgccacaaatgacaa	Int_TFF3_35
cagagagaggagaggagat	Int_TFF3_36
cacgatgtggggcagctaag	Int_TFF3_37
agatggccttatagccaatcg	Int_TFF3_38
tgagctggcacttgacatg	Int_TFF3_39
tggatgctggttcaggaag	Int_TFF3_40
cctaggacaaagggaagag	Int_TFF3_41
catcattgatcaggagccg	Int_TFF3_42
agtggagctgaaggaaacagg	Int_TFF3_43
cactcagaggagagtgtagt	Int_TFF3_44
cagggatcaggcagataaca	Int_TFF3_45
tcgggttgctaagcaagata	Int_TFF3_46
atgggggaatagccaagatt	Int_TFF3_47
acagagcaaaggctgtgtg	Int_TFF3_48

Supp. table 2

ExTFF3 probe sequence

Probe Sequence (5' to 3')	Probe sequence name
aaggggagaggtcagttca	Exn_TFF3_1
ccagacagttttctctcc	Exn_TFF3_2
ctgcatgcctttgtcaagc	Exn_TFF3_3
ttgtttgcttggggaaggc	Exn_TFF3_4
tgtttgcacagctgctctg	Exn_TFF3_5
aggaggcctcatttatgca	Exn_TFF3_6
ctcaggactcgcttcattg	Exn_TFF3_7
cagcatgcagagcgctctg	Exn_TFF3_8
aggacagcaaggccaggac	Exn_TFF3_9
cacgtactcctcagcagag	Exn_TFF3_10
acggcacactggtttgcag	Exn_TFF3_11
actccttgggggtgacatg	Exn_TFF3_12
cagggatcctggatcaaa	Exn_TFF3_13
ggggccttgaacaccaagg	Exn_TFF3_14
gtgcctcagaaggtgcatt	Exn_TFF3_15
caatcacagccgggcaagg	Exn_TFF3_16
gctgagatgaacagtgctt	Exn_TFF3_17
cgggagcaaaaggacagaa	Exn_TFF3_18
acttcagcagaagcgctt	Exn_TFF3_19
gacatcaggctccagatat	Exn_TFF3_20
catgggacctttattcgtt	Exn_TFF3_21
gaagaactgtcctcgggtg	Exn_TFF3_22
acctcagaaaagtctcaggc	Exn_TFF3_23
ccacgacgcagcagaaata	Exn_TFF3_24
cttagggagtcttttctg	Exn_TFF3_25
atccttgcactgcactgcag	Exn_TFF3_26
gtgccagtctggattcaaa	Exn_TFF3_27
ctaggcttctgtgacgt	Exn_TFF3_28
gaagcggcacttacagtg	Exn_TFF3_29
gtatttttctgcttccc	Exn_TFF3_30
ttattggcttgctgtcttc	Exn_TFF3_31
ggaccacttggagacag	Exn_TFF3_32
gtagcgagagtggttgga	Exn_TFF3_33
accaggaatagtagaagt	Exn_TFF3_34
gcctgtttgtcataatgt	Exn_TFF3_35

Supp. table 3

InTFF1 probe sequence

Probe Sequence (5' to 3')	Probe sequence name
acagagcaggaagaagcacg	Int_TFF1_1
tgacacttgggaggattgta	Int_TFF1_2
ccaacactcctctttgaaa	Int_TFF1_3
aaagcactattctgagaccc	Int_TFF1_4
gggagatgttggcatgaaca	Int_TFF1_5
catttgggcctatctggatg	Int_TFF1_6
caggaaacctgtaggagagg	Int_TFF1_7
atgagagaggtggcttgac	Int_TFF1_8
cagggaatatgggttccag	Int_TFF1_9
caaccgatccactgacaac	Int_TFF1_10
ttgagatgcaaacacttccc	Int_TFF1_11
atccattctgcaggtaaagg	Int_TFF1_12
cacacagcatttctgacag	Int_TFF1_13
aaagtgcagtgcagatgc	Int_TFF1_14
aataagttattcagctcccc	Int_TFF1_15
ccaccttgaaactgtaccta	Int_TFF1_16
aacgtagaagccctttctc	Int_TFF1_17
ctctggaaacccttgccttg	Int_TFF1_18
ccacagcagatcaagagg	Int_TFF1_19
agttcgttctgtacaccgag	Int_TFF1_20
aaagacgctctgagccttga	Int_TFF1_21
gtaacaggatcatggacctg	Int_TFF1_22
ttccacacacgaatgcacta	Int_TFF1_23
ctttgtcttgcgcgatgat	Int_TFF1_24
tcaggagggaagatttccag	Int_TFF1_25
tggtgaggagatcatcat	Int_TFF1_26
cactgtgggggtcaagagag	Int_TFF1_27
cgacgtgaaggatgatcatg	Int_TFF1_28
tgggaaggatccgtttcag	Int_TFF1_29
gtctttggagaaagtgttcc	Int_TFF1_30
cttgtctgttgcctatcag	Int_TFF1_31
ataggaggagggaactcag	Int_TFF1_32
agcagggggtgaaagaggag	Int_TFF1_33
taggatctggtggttgacag	Int_TFF1_34
cgcggaatcaaaggcttcag	Int_TFF1_35
tctaaccacagagcttcagg	Int_TFF1_36
cgcaggatcttatgcaagag	Int_TFF1_37
agaggagggaagacacgtgga	Int_TFF1_38
aagcagggcagtgaggaaag	Int_TFF1_39
cagaggggtacaagctgtgaa	Int_TFF1_40
attgcaggtaaggtagagg	Int_TFF1_41
gtgtgttcaggagggtgaaa	Int_TFF1_42
ttcaccagggtgaagatggag	Int_TFF1_43
aggttgaacaagcaggggg	Int_TFF1_44
atgtgcgagacaacgcaagg	Int_TFF1_45
tctgatggaggagaaaggca	Int_TFF1_46
ttgtggacttgcacatttc	Int_TFF1_47
gctcagggaatgcaaggt	Int_TFF1_48

Supp. table 4

ExTFF1 probe sequence

Probe Sequence (5' to 3')	Probe sequence name
gcgaccccgagtcagggat	Exn_TFF1_1
tgccctcctctctgctcaa	Exn_TFF1_2
cttggttctccatgggtggcc	Exn_TFF1_3
caggaccagggcgagatc	Exn_TFF1_4
gccgagggccagcatggac	Exn_TFF1_5
tgtctgggcctcggccagg	Exn_TFF1_6
ggggccactgtacacgtct	Exn_TFF1_7
ccacaattctgtctttcac	Exn_TFF1_8
gagggcgtgacaccaggaa	Exn_TFF1_9
cagcccttattgcacact	Exn_TFF1_10
cacgaacggtgtcgtcgaa	Exn_TFF1_11
gatagaagcaccaggggac	Exn_TFF1_12
gagggacgtcgatggatt	Exn_TFF1_13
tcgtataaaaaggccatac	Exn_TFF1_14
attctaattcttcagaacc	Exn_TFF1_15
gtcttaaatgacttttcta	Exn_TFF1_16
atgctgatcagagcctctg	Exn_TFF1_17
tgtgtaaaggcatagctgg	Exn_TFF1_18
caccactggcgccgtgac	Exn_TFF1_19
gactcaggctaccccattg	Exn_TFF1_20
ttcctggacctgaatgcag	Exn_TFF1_21
tccttagccctgecttfc	Exn_TFF1_22
gtctaaaattcacactcct	Exn_TFF1_23
caggcagatccctgcagaa	Exn_TFF1_24
ggacggcaccgcgtcagga	Exn_TFF1_25
tgggactaatcaccgtgct	Exn_TFF1_26
ggtggagggtggcagccgag	Exn_TFF1_27
agaagcgtgtctgaggtgt	Exn_TFF1_28
tgtgagccgaggcacagct	Exn_TFF1_29
gtcagagcagtcattctgt	Exn_TFF1_30
ggccaattttgagtagtca	Exn_TFF1_31
atcgatctcttttaatttt	Exn_TFF1_32

Supp. table 5

4C Seq oligo sequences

Sequence name	Sequence (5' to 3')
4C <i>TFF1</i> Enhancer barcode1 F	AATGATACGGCGACCACCGAGATCTACACTCTTCCCTACACGACG CTCTCCGATCTAGAAGTCCTTGAGTGGGGAGATGATC
4C <i>TFF1</i> Enhancer barcode2 F	AATGATACGGCGACCACCGAGATCTACACTCTTCCCTACACGACG CTCTCCGATCTAGCGACCCTTGAGTGGGGAGATGATC
4C <i>TFF1</i> Enhancer barcode3 F	AATGATACGGCGACCACCGAGATCTACACTCTTCCCTACACGACG CTCTCCGATCTCCAATTCCTTGAGTGGGGAGATGATC
4C <i>TFF1</i> R	CAAGCAGAAGACGGCATACGAGATAGGTTCGAGTGACTGGAGTTC AGACGTGTGCTCTTCCGATCTCCGCTGTTGCGACACACCAGAGAGC

Supp. table 6

qPCR oligo sequences

Sequence name	Sequence (5' to 3')
<i>TFF1</i> US F	GCCCTTATTTGCACACTGGG
<i>TFF1</i> US R	CATCAAGGAATTCAGCCAC
<i>TFF3</i> US F	CAATCACAGCCGGCAAGGGT
<i>TFF3</i> US R	GGGAGGAGGCAGCACTAGGT
GAPDH F	CGCTCTGCTCCTCTGTT
GAPDH R	CCATGGTGTCTGAGCGATGT

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Reviewer #1 (Public review):

Summary:

The manuscript by Bohra et al. describes the indirect effects of ligand-dependent gene activation on neighboring non-target genes. The authors utilized single-molecule RNA-FISH (targeting both mature and intronic regions), 4C-seq, and enhancer deletions to demonstrate that the non-enhancer-targeted gene TFF3, located in the same TAD as the target gene TFF1, alters its expression when TFF1 expression declines at the end of the estrogen signaling peak. Since the enhancer does not loop with TFF3, the authors conclude that mechanisms other than estrogen receptor or enhancer-driven induction are responsible for TFF3 expression. Moreover, ER α intensity correlations show that both high and low levels of ER α are unfavorable for TFF1 expression. The ER α level correlations are further supported by overexpression of GFP-ER α . The authors conclude that transcriptional machinery used by TFF1 for its acute activation can negatively impact the TFF3 at peak of signaling but once, the condensate dissolves, TFF3 benefits from it for its low expression.

Strengths:

The findings are indeed intriguing. The authors have maintained appropriate experimental controls, and their conclusions are well-supported by the data.

Weaknesses:

There are some major and minor concerns that related to approach, data presentation and discussion. But the authors have greatly improved the manuscript during the revision work.

Comments on latest version:

The authors have done a lot of work for the revision. The manuscript has been greatly improved.

<https://doi.org/10.7554/eLife.102417.2.sa2>

Reviewer #3 (Public review):

Summary:

In this manuscript Bohra et al. measure the effects of estrogen responsive gene expression upon induction on nearby target genes using a TAD containing the genes TFF1 and TFF3 as a model. The authors propose that there is a sort competition for transcriptional machinery between TFF1 (estrogen responsive) and TFF3 (not responsive) such that when TFF1 is activated and machinery is recruited, TFF3 is activated after a time delay. The authors attribute this time delay to transcriptional machinery that was being sequestered at TFF1 becomes available to the proximal TFF3 locus. The authors demonstrate that this activation is not dependent on contact with the TFF1 enhancer through deletion, instead they conclude that it is dependent on a phase-separated condensate which can sequester transcriptional machinery. Although the manuscript reports an interesting observation that there is a dose dependence and time delay on the expression of TFF1 relative to TFF3, there is much room for improvement in the analysis and reporting of the data. Most importantly there is no direct test of condensate formation at the locus in the context of this study: i.e. dissolution upon the enhancer deletion, decay in a temporal manner, and dependence of TFF1 expression on condensate formation. Using 1,6' hexanediol to draw conclusion on this matter is not adequate to draw conclusions on the effect of condensates on a specific genes activity given current knowledge on its non-specificity and multitude of indirect effects. Thus, in my opinion the major claim that this effect of a time delayed expression of TFF3 being dependent on condensates is not supported by the current data.

Strengths:

The dependence of TFF1 expression on a single enhancer and the temporal delay in TFF3 is a very interesting finding.

The non-linear dependence of TFF1 and TFF3 expression on ER concentration is very interesting with potentially broader implications.

The combined use of smFISH, enhancer deletion, and 4C to build a coherent model is a good approach.

Weaknesses:

There is no direct observation of a condensate at the TFF1 and TFF3 locus and how this condensate changes over time after E2 treatment, upon enhancer deletion, whether transcriptional machinery is indeed concentrated within it, and other claims on condensate function and formation made in the manuscript. The use of 1,6' HD is not appropriate to test this idea given how broadly it acts.

Comments on latest version:

I don't think the response to Reviewer 2's comment on LLPS condensates on TFF1 are adequate and given this point is essential to the claims of the manuscript they must be addressed. Namely, the data from Saravanan, 2020 actually suggest that condensate formation at the locus is not very predictive and barely enriched over random spots. The claims in the manuscript on the dependence of the condensate being responsible for sequestering transcriptional machinery are quite strong and the crux of the current model. To continue to make this claim (which I don't think is necessary since there are other possible models) the authors must test if the condensate at this locus (1) shows time dependent behavior, (2) is not present or weakened at the locus in cells that show high TFF3 expression, (3) is indeed enriched for transcriptional machinery when TFF1 peaks. The use of 1,6 hexanediol is not appropriate as pointed out by reviewer 2 and is no longer considered as an appropriate experiment by many as the whole notion of LLPS forming nuclear condensates is now under question. Such condensates can form through a variety of mechanisms as reviewed for example by Mittal and Pappu (A conceptual framework for understanding phase separation and addressing open questions and challenges, Molecular Cell, 2022). Furthermore, given the distance between TFF1 and TFF3 it is hard to imagine that if a condensate that concentrates machinery in a non-stoichiometric manner was forming how it would not boost expression on both genes and be just specific to one. There must be another mechanism in my opinion.

I would recommend the authors remove this aspect of their manuscript/model and simply report their interesting findings that are actually supported by data: The temporal delay of TFF3 expression, the dependence on ER concentration, and the enhancer dependence.

<https://doi.org/10.7554/eLife.102417.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1:

Summary:

The manuscript by Bohra et al. describes the indirect effects of ligand-dependent gene activation on neighboring non-target genes. The authors utilized single-molecule RNA-FISH (targeting both mature and intronic regions), 4C-seq, and enhancer deletions to

demonstrate that the non-enhancer-targeted gene TFF3, located in the same TAD as the target gene TFF1, alters its expression when TFF1 expression declines at the end of the estrogen signaling peak. Since the enhancer does not loop with TFF3, the authors conclude that mechanisms other than estrogen receptor or enhancer-driven induction are responsible for TFF3 expression. Moreover, ERα intensity correlations show that both high and low levels of ERα are unfavorable for TFF1 expression. The ERα level correlations are further supported by overexpression of GFP-ERα. The authors conclude that transcriptional machinery used by TFF1 for its acute activation can negatively impact the TFF3 at peak of signaling but once, the condensate dissolves, TFF3 benefits from it for its low expression.

Strengths:

The findings are indeed intriguing. The authors have maintained appropriate experimental controls, and their conclusions are well-supported by the data.

Weaknesses:

There are some major and minor concerns that related to approach, data presentation and discussion. But I think they can be fixed with more efforts.

We thank the reviewer for their positive comments on the paper. We have addressed all their specific recommendations below.

The deletion of enhancer reveals the absolute reliance of TFF1 on its enhancers for its expression. Authors should elaborate more on this as this is an important finding.

We thank the reviewer for the comment. We have now added a more detailed discussion on the requirement of enhancer for TFF1 expression in the revised manuscript (line 368-385).

In Fig. 1, TFF3 expression is shown to be induced upon E2 signaling through qRT-PCR, while smFISH does not display a similar pattern. The authors attribute this discrepancy to the overall low expression of TFF3. In my opinion, this argument could be further supported by relevant literature, if available. Additionally, does GRO-seq data reveal any changes in TFF3 expression following estrogen stimulation? The GRO-seq track shown in Fig.1 should be adjusted to TFF3 expression to appreciate its expression changes.

We have now included a browser shot image of TFF3 region showing GRO-Seq signal at E2 time course (Fig. S1C). We observed an increased transcription towards the 3' end of TFF3 gene body at 3h. The increased transcription at 3h, corroborates with smFISH data. The relative changes of TFF3 expression measured by qRT-PCR and smFISH for intronic transcripts are somewhat different, we speculate that such biased measurements that are dependent on PCR amplifications could be more for genes that express at low levels and smFISH using intronic probes may be a more sensitive assay to detect such changes.

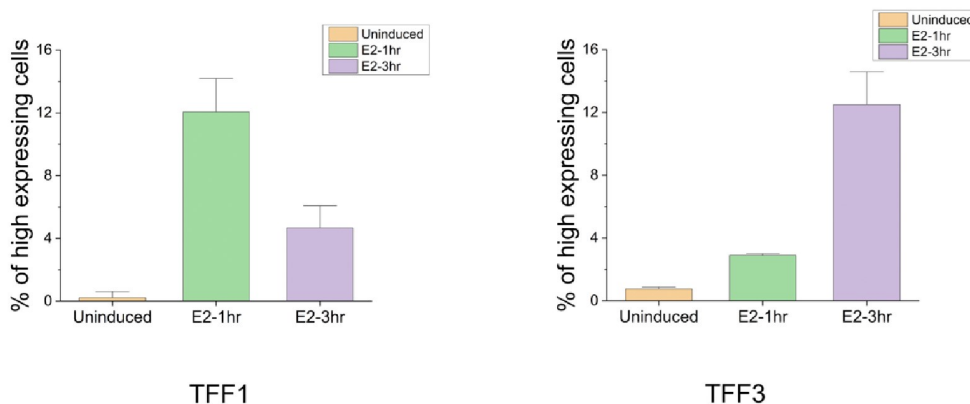
Since the mutually exclusive relationship between TFF1 and TFF3 is based on snap shots in fixed cells, can authors comment on whether the same cell that expresses TFF1 at 1h, expresses TFF3 at 3h? Perhaps, the calculations taking total number of cells that express these genes at 1 and 3h would be useful.

Like pointed out by the reviewer, since these are fixed cells, we cannot comment on the fate of the same cell at two time points. To further address this limitation, future work could employ cells with endogenous tags for TFF1 and TFF3 and utilize live cell imaging techniques. In a fixed cell assay, as the reviewer suggests, it can be investigated whether a similar fraction shows high TFF3 expression at 3h, as the fraction that shows high TFF1 expression at 1 h. To

quantify the fractions as suggested by the reviewer, we plotted the fraction of cells showing high TFF1 and TFF3 expression at 1h and 3h. We identify truly high expressing cells by taking mean and one standard deviation (for single cell level data) at E2-1hr as the threshold for TFF1 (80 and above transcript counts) and mean and one standard deviation (for single cell level data) at E2-3hr as the threshold for TFF3 (36 and above transcript counts). The fraction with high TFF1 expression at 1h (12.06 ± 2.1) is indeed comparable to that with high TFF3 expression at 3h (12.50 ± 2.0) (Fig. 2C and Author response image 1). We should note that if the transcript counts were normally distributed, a predetermined fraction would be expected to be above these thresholds and comparable fractions can arise just from underlying statistics. But in our experiments, this is unlikely to be the case given the many outliers that affect both the mean and the standard deviation, and the lack of normality and high dispersion in single cell distributions. Of course, despite the fractions being comparable, we cannot be certain if it is the same set of cells that go from high expression of TFF1 to high expression of TFF3, but definitely that is a possibility. We thank the reviewer for pointing out this comparison.

Author response image 1.

The graph represents the percent of cells that show high expression for TFF1 and TFF3 at 1h and 3h post E2 signaling. The threshold was collected by pooling in absolute RNA counts from 650 analyzed cells (as in Fig. 2C). The mean and standard deviation over single cell data were calculated. Mean plus one standard deviation was used to set the threshold for identifying high expressing cells. For TFF1, as it maximally expresses at 1h the threshold used was 80. For TFF3, as it maximally expresses at 3h the threshold used was 36. Fraction of cells expressing above 80 and 36 for TFF1 and TFF3 respectively were calculated from three different repeats. Mean of means and standard deviations from the three experiments are plotted here.



Authors conclude that TFF3 is not directly regulated by enhancer or estrogen receptor. Does ERα bind on TFF3 promoter?

The ERα ChIP-seq performed at 1h and 3h of signaling suggests that TFF3 promoter is not bound by ERα as shown in supplementary Fig. 1B and S1B. However, one peak upstream to TFF1 promoter is visible and that is lost at 3h.

Minor comments:

Reviewer's comment -The figures would benefit from resizing of panels. There is very little space between the panels.

We have now resized the figures in the revised manuscript.

The discussion section could include an extrapolation on the relationship between ERα concentration and transcriptional regulation. Given that ERα levels have been shown to play a critical role in breast cancer, exploring how varying concentrations of ERα affect gene expression, including the differential regulation of target and non-target genes, would provide valuable insights into the broader implications of this study.

This is a very important point that was missing from the manuscript. We have included this in the discussion in the revised manuscript (line 426-430).

Reviewer #2:

Summary:

In this manuscript by Bohra et al., the authors use the well-established estrogen response in MCF7 cells to interrogate the role of genome architecture, enhancers, and estrogen receptor concentration in transcriptional regulation. They propose there is competition between the genes TFF1 and TFF3 which is mediated by transcriptional condensates. This reviewer does not find these claims persuasive as presented. Moreover, the results are not placed in the context of current knowledge.

Strengths:

High level of ERα expression seems to diminish the transcriptional response. Thus, the results in Fig. 4 have potential insight into ER-mediated transcription. Yet, this observation is not pursued in great depth however, for example with mutagenesis of ERα. However, this phenomenon - which falls under the general description of non monotonic dose response - is treated at great depth in the literature (i.e. PMID: 22419778). For example, the result the authors describe in Fig. 4 has been reported and in fact mathematically modeled in PMID 23134774. One possible avenue for improving this paper would be to dig into this result at the single-cell level using deletion mutants of ERα or by perturbing co-activators.

We thank the reviewer for pointing us to the relevant literature on our observation which will enhance the manuscript. We have discussed these findings in relations to ours in the discussion section (Line 400-413). We thank the reviewer for insight on non-monotonic behavior.

Weaknesses:

There are concerns with the sm-RNA FISH experiments. It is highly unusual to see so much intronic signal away from the site of transcription (Fig. 2) (PMID: 27932455, 30554876), which suggests to me the authors are carrying out incorrect thresholding or have a substantial amount of labelling background. The Cote paper cited in the manuscript is likewise inconsistent with their findings and is cited in a misleading manner: they see splicing within a very small region away from the site of transcription.

We thank the reviewer for this comment, and apologize if they feel we misrepresented the argument from Cote et al. This has now been rectified in the manuscript. However, we do not agree that the intronic signals away from the site of transcription are an artefact. First, the images presented here are just representative 2D projections of 3D Z-stacks; whereas the full 3D stack is used for spot counting using a widely-used algorithm that reports spot counts that are constant over wide range of thresholds (Raj et al., 2008). The veracity of automated counts

was first verified initially by comparison to manual counts. Even for the 2D representations the extragenic intronic signals show up at similar thresholds to the transcription sites.

The signal is *not* non-specific arising from background labeling, explained by following reasons:

- To further support the time-course smFISH data and its interpretation without depending on the dispersed intronic signal, we have analyzed the number of alleles firing/site of transcription at a given time in a cell under the three conditions. We counted the sites of transcription in a given cell and calculated the percentage of cells showing 1,2,3,4 or >4 sites. We see that the percent of cells showing a single site of transcription for TFF1 is very high in uninduced cells and this decreases at 1h. At 1h, the cells showing 2, 3 and 4 sites of transcription increase which again goes down at 3h (Author response image 2A). This agrees with the interpretation made from mean intronic counts away from the site of transcription. Similarly, for TFF3, the number of cells showing 2,3 and 4 sites of transcription increase slightly at 3hr compared to uninduced and 1hr (Author response image 2B). We can also see that several cells have no alleles firing at a given time as has been quantified in the graphs on right showing total fraction of cells with zero versus non-zero alleles firing (Author response image 2A-B). A non-specific signal would be present in all cells.
- There is literature on post-transcriptional splicing of RNA beyond our work, which suggests that intronic signal can be found at relatively large distances away from the site of transcription. Waks et al. showed that some fraction of unspliced RNA could be observed up to 6-10 microns away from the site of transcription suggesting that there can be a delay between transcription and (alternative) splicing (Waks et al., 2011). Pannuclear disperse intronic signals can arise as there can be more than one allele firing at a time in different nuclear locations. The spread of intronic transcripts in our images is also limited in cells in which only 1 allele is firing at E2-1 hour (Author response image 2C) or uninduced cells (Author response image 2D). Furthermore, Cote et al. discuss that “Of note, we see that increased transcription level correlates with intron dispersal, suggesting that the percentage of splicing occurring away from the transcription site is regulated by transcription level for at least some introns. This may explain why we observe posttranscriptional splicing of all genes we measured, as all were highly expressed.” This is in line with our interpretation that intron signal dispersal can occur in case of posttranscriptional splicing (Coté et al., 2023). Additionally, other studies have suggested that transcripts in cells do not necessarily undergo co-transcriptional splicing which leads us to conclude that intronic signal can be found farther away from the site of transcription. Coulon et al. showed that splicing can occur after transcript release from the site and suggested that no strict checkpoint exists to ensure intron removal before release which results in splicing and release being kinetically uncoupled from each other (Coulon et al., 2014). Similarly, using live-cell imaging, it was shown that splicing is not always coupled with transcription, and this could depend on the nature and structural features of transcript (such as blockage of polypyrimidine tract which results in delayed recognition) (Vargas et al., 2011). Drexler et al. showed that as opposed to drosophila transcripts that are shorter, in mammalian cells, splicing of the terminal intron can occur post-transcriptionally (Drexler et al., 2020). Using RNA polymerase II ChIP-Seq time course data from ERα activation in the MCF-7 cells, Honkela et al. showed that large number of genes can show significant delays between the completion of transcription and mRNA production (Honkela et al., 2015). This was attributed to faster transcription of shorter genes which results in splicing delays suggesting rapid completion of transcription on shorter genes can lead to splicing-associated delays (Honkela et al., 2015). More recently, comparisons of nascent and mature RNA levels suggested a time lapse between transcription and splicing for the genes that are early responders during signaling (Zambrano et al., 2020). The presence of significant numbers of TFF1 nascent RNA in the nucleus in our data corroborates with above observations.

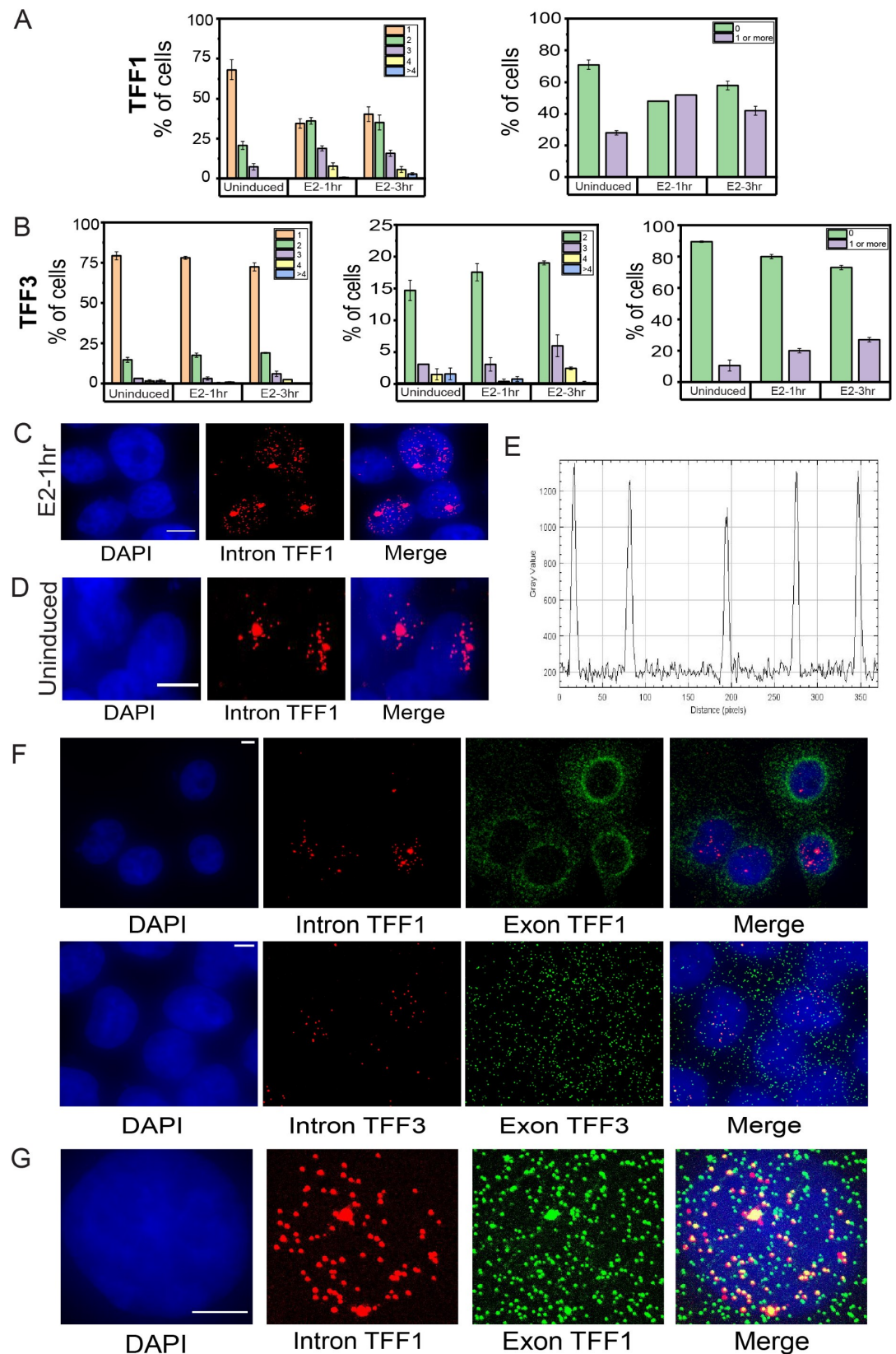
- Uniform intensities across many transcripts suggests these are true signal arising from RNA molecules which would not be the case for non-specific, background signal (Author response image 2E).
- Splicing occurs in the nucleus and intron containing pre-transcripts should be nuclear localized. Thus, intronic signals should remain localized to the nucleus unlike the mature mRNA which translocate to the cytoplasm after processing and thus exonic signals can be found both in the nucleus and the cytoplasm. In keeping with this, we observe no signal in the cytoplasm for the intronic probes and it remains localized within the nucleus as expected and can be seen in Author response image 2F, while exonic signals are observed in both compartments. This suggests to us that the signal is coming from true pre-transcripts. There is no reason for non-specific background labelling to remain restricted to the nucleus.
- We observe that the mean intronic label counts for both the genes *TFF1* and *TFF3* increases upon E2-induction compared to uninduced condition (Fig. 2B). Similarly, the mean intronic count for both genes reduce drastically in the *TFF1*-enhancer deleted cells (Fig. 3C, D). This change in the number of intronic signal specifically on induction and enhancer deletion suggests that the signal is not an artefact and arises from true nascent transcripts that are sensitive to stimulus or enhancer deletion.
- We expect colocalization of intronic signal with exonic signals in the nucleus, while there can be exonic signals that do not colocalize with intronic, representing more mature mRNA. Indeed, we observe a clear colocalization between the intronic and exonic signals in the nucleus, while exonic signals can occur independent of intronic both in the nucleus and the cytoplasm. This clearly demonstrates that the intronic signals in our experiments are specific and not simply background labelling (Author response image 2G).

These studies and the arguments above lead us to conclude that the presence of intronic transcripts in the nucleus, away from the site of transcription is not an artefact. We hope the reviewer will agree with us. These analyses have now been included in the manuscript as Supplementary Figure 6 and have been added in the manuscript at line numbers 106-111, 201204, 215-217 and line 231-235. We thank the reviewer for raising this important point.

Author response image 2.

Dynamic induction and RNA localization of *TFF1* and *TFF3* transcription across cell populations using smRNA FISH A. Bar graph depicting the percentage of cells with 1,2,3,4, or greater than 4 sites of transcription for *TFF1* (left) is shown. The graph shows the mean of means from different repeats of the experiment, and error bars denote SEM ($n > 200$, $N = 3$). Only the cells with at least one allele firing were counted and cells with no alleles were not included in this. The graph on right shows the number of cells with zero or non-zero number of alleles firing. B. Bar graph depicting the percentage of cells with 1,2,3,4 or greater than 4 sites of transcription for *TFF3* (left) is shown. The graph shows the mean of means from different repeats of the experiment, and error bars denote SEM ($n > 200$, $N = 3$). Only the cells with at least one allele firing were counted and cells with no alleles were not included in this. The graph in the middle shows the number of cells with 2,3,4 or greater than 4 sites of transcription for *TFF3*. The graph on the right shows the number of cells with zero or non-zero number of alleles firing. C. Images from single molecule RNA FISH experiment showing transcripts for *InTFF1* in cells induced for 1 hour with E2. The image shows that when a single allele of *TFF1* is firing, the transcripts show a more spatially restricted localisation. The scale bar is 5 microns. D. Images from single molecule RNA FISH experiment showing transcripts for *InTFF1* in uninduced cells. The image shows that when a single allele of *TFF1* is firing and transcription is low, the transcripts show a more spatially restricted localisation. The scale bar is 5 microns. E. Line profile through several transcripts in the nucleus show uniform and similar intensities indicating that these are true signals. F. 60X Representative

images from a single molecule RNA FISH experiment showing transcripts for InTFF1 and ExTFF1 (top) and InTFF3 and ExTFF3 (bottom). The image shows that there is no intronic signal in the cytoplasm, while exonic signals can be found both in the nucleus and the cytoplasm. The scale bar is 5 microns. G. 60X Representative images from single molecule RNA FISH experiment showing transcripts for InTFF1 and ExTFF1. The image shows that all intronic signals are colocalized with exonic signals, but all exonic signals are expectedly not colocalized with intronic signals, representing more mature mRNA. The scale bar is 5 microns.



One substantial way to improve the manuscript is to take a careful look at previous single cell analysis of the estrogen response, which in some cases has been done on the exact same genes (PMID: 29476006, 35081348, 30554876, 31930333). In some of these

cases, the authors reach different conclusions than those presented in the present manuscript. Likewise, there have been more than a few studies that have characterized these enhancers (the first one I know of is: PMID 18728018). Also, Oh et al. 2021 (cited in the manuscript) did show an interaction between TFF1e and TFF3, which seems to contradict the conclusion from Fig. 3. In summary, the results of this paper are not in dialogue with the field, which is a major shortcoming.

We thank the reviewer for pointing out these important studies. The studies from Prof. Larson group are particularly very insightful (Rodriguez et al., 2019). We have now included this in the discussion (line 106-111 and line 420-424) where we suggest the differences and similarities between our, Larson's group and also Mancini's group (Patange et al., 2022; Stossi et al., 2020).

The 4C-Seq data from the manuscript Oh et al. 2021 is exactly consistent with our observation from Fig 3 as they also observed little to no interaction between TFF1e and TFF3p in WT cells, only upon TFF1p deletion, did the TFF1e become engaged with the TFF3p. In agreement with this, we also observe little to no interaction between TFF1e and TFF3p in WT cells (Fig.3A). This is also consistent with our competition model for resources between these two genes. Oh et al. shows interaction between TFF1e and TFF3 when the TFF1 promoter is deleted showing that when the primary promoter is not available the enhancer is retargeted to the next available gene (Oh et al., 2021). It does not show that in WT or at any time point of E2 signalling does TFF1e and TFF3 interact.

In the opinion of this reviewer, there are few - if any - experiments to interrogate the existence of LLPS for diffraction-limited spots such as those associated with transcription. This difficulty is a general problem with the field and not specific to the present manuscript. For example, transient binding will also appear as a dynamic 'spot' in the nucleus, independently of any higher-order interactions. As for Fig. 5, I don't think treating cells with 1,6 hexanediol is any longer considered a credible experiment. For example, there are profound effects on chromatin independent of changes in LLPS (PMID: 33536240).

We are cognizant of and appreciate the limitations pointed out by the reviewer. We and others have previously shown that ERa forms condensates on TFF1 chromatin region using ImmunoFISH assay (Saravanan et al., 2020). The data below shows the relative mean ERa intensity on TFF1 FISH spots and random regions clearly showing an appearance of the condensate at the TFF1 site. Further, the deletion of TFF1e causes the reduction in size of this condensate. Thus, we expect that these ERa condensates are characterized by higher-order interactions and become disrupted on treatment with 1,6-hexanediol. These condensates are the size of below micron as mentioned by the reviewer, but most TF condensates are of the similar sizes. We agree with the reviewer that 1,6- hexanediol treatment is a brute-force experiment with several irreversible changes to the chromatin. Although we have tried to use it at a low concentration for a short period of time and it has been used in several papers (Chen et al., 2023; Gamliel et al., 2022). The opposite pattern of TFF1 vs. TFF3 expression upon 1,6- hexanediol treatment suggests that there is specificity. Further, to perturb condensates, mutants of ERa can be used (N-terminus IDR truncations) however, the transcriptional response of these mutants is also altered due to perturbed recruitment of coactivators that recognize Nterminus of ER, restricting the distinction between ERa functions and condensate formation.

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